

Ozone Production in the Rural Troposphere and the Implications for Regional and Global Ozone Distributions

S. C. LIU, M. TRAINER,¹ F. C. FEHSENFELD, D. D. PARRISH,^{1, 2} E. J. WILLIAMS,¹
D. W. FAHEY, G. HÜBLER,¹ AND P. C. MURPHY¹

Aeronomy Laboratory, Environmental Research Laboratories, NOAA, Boulder, Colorado

The relationship between O_3 and NO_x ($NO + NO_2$) which was measured during summer and winter periods at Niwot Ridge, Colorado, has been analyzed and compared to model calculations. Both model calculations and observations show that the daily O_3 production per unit of NO_x is greater for lower NO_x . Model calculations without nonmethane hydrocarbons (NMHC) tend to underestimate the O_3 production rate at NO_x higher than 1.5 parts per billion by volume and show the opposite dependence on NO_x . The model calculations with NMHC are consistent with the observed data in this regime and demonstrate the importance of NMHC chemistry in the O_3 production. In addition, at eight other rural stations with concurrent O_3 and NO_x measurements in the central and eastern United States the daily O_3 increase in summer also agrees with the O_3 and NO_x relationship predicted by the model. The consistency of the observed and model-calculated daily summer O_3 increase implies that the average O_3 production in rural areas can be predicted if NO_x is known. The dependence of O_3 production rate on NO_x deduced in this study provides the basis for a crude estimate of the total O_3 production. For the United States an average summer column O_3 production of about $1 \times 10^{12} \text{ cm}^{-2} \text{ s}^{-1}$ from anthropogenically emitted NO_x and NMHC is estimated. This photochemical production is roughly 20 times the average cross-tropopause O_3 flux. Production of O_3 from NO_x that is emitted from natural sources in the United States is estimated to range from 1.9×10^{11} to $12 \times 10^{11} \text{ cm}^{-2} \text{ s}^{-1}$, which is somewhat smaller than ozone production from anthropogenic NO_x sources. Extrapolation to the entire northern hemisphere shows that in the summer, 3 times as much O_3 is generated from natural precursors as those of anthropogenic origin. The winter daily O_3 production rate was found to be about 10% of the summer value at the same NO_x level. However, because of longer NO_x lifetime in the winter, the integrated O_3 production over the lifetime of NO_x may be comparable to the summer value. Moreover, because the natural NO_x sources are substantially smaller in the winter, the wintertime O_3 budget in the northern hemisphere should be dominated by ozone production from anthropogenic ozone precursors. The photochemical lifetime of O_3 in the winter in the mid-latitude is approximately 200 days. We propose that this long lifetime allows anthropogenically produced O_3 to accumulate and contribute substantially to the observed spring maximum that is usually attributed to stratospheric intrusion. Furthermore, the anthropogenic O_3 may be transported not only zonally but also to lower latitudes. Thus the long-term interannual increase in O_3 , observed in the winter and spring seasons at Mauna Loa, may be due to the same anthropogenic influences as the similar winter trend observed at Hohenpeissenberg, Germany.

INTRODUCTION

Since the initial prediction of an active hydrogen radical photochemistry in the natural troposphere by Levy [1971], the photochemical production and loss of tropospheric ozone have been investigated extensively. By analogy to the urban ozone formation mechanism, Crutzen [1973] and Chameides and Walker [1973] argued that photochemical production of ozone in the troposphere is much greater than the flux from the stratosphere. Later advances in the knowledge of the tropospheric distribution of nitrogen oxides [e.g., Noxon 1978; Kley *et al.*, 1981] resulted in an improved understanding of the ozone budget [Fishman *et al.*, 1979; Liu *et al.*, 1980; Logan *et al.*, 1981; Crutzen and Gidel, 1983]. These studies generally confirmed the earlier calculations. The predicted production and loss rates were smaller, but the net production of ozone in the troposphere still remained a few times the cross-tropopause flux of ozone from the stratosphere.

The studies cited above are model evaluations of global production and loss of O_3 based on limited knowledge of the distribution and budget of NO_x . Direct observation of photochemical production and loss of O_3 and its dependence on NO_x is required to validate such model predictions. Some indirect evidence for photochemical production is available [Fishman *et al.*, 1979; Fishman and Seiler, 1983], but it is not fully quantitative and may be subject to other interpretations [Liu *et al.*, 1980; Logan, 1985]. On the other hand, observation of NO_x mixing ratios less than 0.01 parts per billion by volume (ppbv) in the mid-Pacific provides evidence for photochemical destruction of ozone in the remote troposphere [Liu *et al.*, 1983].

Extensive data on O_3 and its precursors have been gathered at several rural stations [Fehsenfeld *et al.*, 1983; Kelly *et al.*, 1984a; Parrish *et al.*, 1986a]. These data allow detailed analysis of the production and loss of O_3 and the relationship of these processes with NO_x and hydrocarbon precursors [Fehsenfeld *et al.*, 1983; Kelly *et al.*, 1984a; Greenberg and Zimmerman, 1984].

In order to evaluate these data, a chemical modeling approach is presented that treats the influence of the combined effect of NO_x and nonmethane hydrocarbons (NMHC), as well as CO and CH_4 , on ozone production. Approximate methods to compensate for the effects of transport and dilution are developed. This treatment provides estimates of ozone

¹ Also at Cooperative Institute for Research in Environmental Sciences, University of Colorado, Boulder.

² Also at Department of Chemistry, Metropolitan State College, Denver, Colorado.

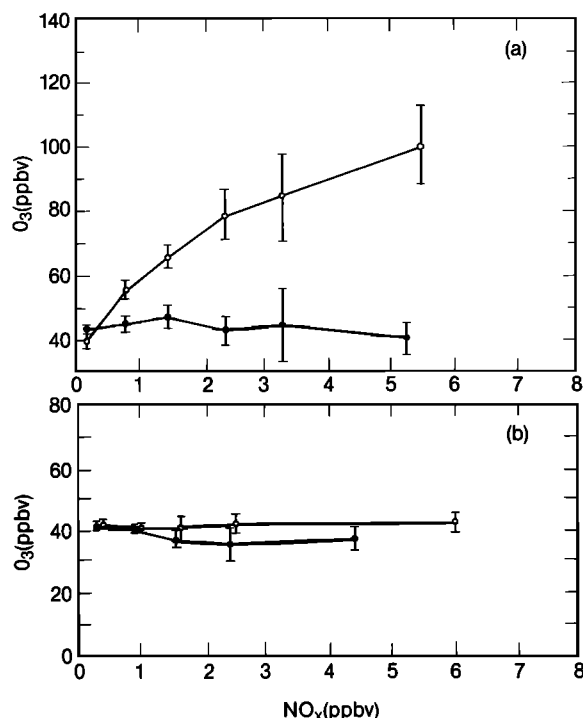


Fig. 1. (a) Summertime (June 1 to August 31) O₃ and NO_x relationship observed under clear sky conditions at Niwot Ridge, Colorado. Solid circles are observed values from 0700 to 1100 MST and open circles are observed values between 1400 and 1900 MST. The vertical bars give the 95% confidence limits for the average. (b) Same as Figure 1a except for winter (December 1 to February 28).

production occurring in an air mass. The model predictions are compared to the diurnal variations of ozone as a function of NO_x mixing ratio measured at Niwot Ridge, Colorado. These results, in turn, are compared with the measured summer ozone increase observed by Kelly *et al.* [1984a] and Research Triangle Institute [1975]. An algorithm is developed to approximate the relative emission of NMHC and NO_x from anthropogenic sources. The model-predicted ozone production as a function of NO_x level is then used to estimate photochemical ozone production associated with natural and anthropogenic NO_x sources as a function of season on regional and global scales.

MEASUREMENTS

The measurement site, instruments, and data were described in detail previously [Fehsenfeld *et al.*, 1983; Parrish *et al.*, 1986a]. A brief summary is given here to facilitate later discussion.

The measurement site is located in a forest clearing in the Rocky Mountains approximately 60 km northwest of metropolitan Denver, Colorado. The site has an elevation of 3.05 km. The prevailing winds are from the west, which bring in clean air; however, there are frequent easterlies (i.e., wind direction is from the east) that transport pollutants from the metropolitan area to the site. As a result, the measurements show large variations in the concentrations of anthropogenic pollutants. Atmospheric trace species were measured concurrently at the site during several extended periods from 1981 to 1984. A large data base of simultaneous measurements of O₃, NO, NO₂, H₂O, UV radiation flux, and meteorological parameters was obtained. CO, CH₄, NMHC, SO₂, particulate

NO₃⁻ and SO₄²⁻ were measured less frequently. Of particular interest were the NMHC which were measured at this site by Roberts *et al.* [1983, 1984, 1985] and Greenberg and Zimmerman [1984]. Only average values of NMHC for the summer and winter are given by the latter investigators. For the hydrocarbons measured by both groups the results were consistent with each other.

Figure 1 shows the O₃ mixing ratios measured at the site in the morning and afternoon during the summer and winter. The data shown include all measurements from June 1 through August 31 of 1981, 1983, and 1984 (Figure 1a) and December 1 through February 28, 1981 (Figure 1b). The open circles represent the observed average values of O₃ within a NO_x interval centered on the symbol for the afternoon between 1400 and 1900 MST. The solid circles give the morning measurements that were made between 0700 and 1100 MST. The vertical bars are 95% confidence levels of the average values. The confidence levels are relatively large for NO_x levels greater than 2 ppbv due to the sparseness of the data for these infrequent polluted levels.

Since the model described below includes no cloud effects, we have excluded data in Figure 1 that were obtained during periods when the photolysis rate of NO₂ was below $2 \times 10^{-3} \text{ s}^{-1}$, i.e., about 20% of the noontime clear sky value [c.f. Parrish *et al.*, 1983]. This is not the best way to screen out data of cloudy days because this criterion is based on only the UV flux measured at the site, which may not always represent general sky conditions accurately. Nevertheless, this criterion is useful for excluding data from heavily overcast skies. The difference between the morning and afternoon curves represents the net daily ozone change. Except at very low NO_x, there is a net increase in O₃ during the day in summertime. It will be shown later that most of the increase is due to photochemical O₃ production (see also Fehsenfeld *et al.* [1983]). Little or no such increase is observed in wintertime.

MODEL CALCULATIONS

The net daily ozone change, indicated below as Q , is the result of the combined processes responsible for ozone production P , loss (including photochemical loss and surface deposition) L , and transport T ,

$$Q = P - L + T \quad (1)$$

where the units are parts per billion by volume (ppbv) of O₃ per day. In order to compare the observed Q to theoretical predictions, model calculations that include the dominant production and loss processes have been made. The calculations use ambient conditions that are appropriate to Niwot Ridge. The model is an extension of the box model described by Liu *et al.* [1980]. This model in its original form neglects transport and surface deposition. The reaction rates have been updated according to the Jet Propulsion Laboratory [1985] recommendations. Table 1 gives a list of the reactions and rate constants included in the model. Reaction schemes for NMHC are adopted from Atkinson *et al.* [1982] and Atkinson and Lloyd [1984]. For natural NMHC the reactions of isoprene have been included that are based on the reaction schemes developed by Lloyd *et al.* [1984]. To study the influence of NMHC on the ozone production, the model calculations were made first excluding then including the observed NMHC mixing ratios.

The distributions of trace gases are controlled, at least in

part, by transport. Since the present model neglects the transport processes, the concentrations of long-lived species that are most sensitive to transport are fixed at their observed values at this site. These include CH_4 , CO , HNO_3 , and H_2O . Because there is significant photochemical production or loss, the concentrations of NO_x , NMHC, and O_3 are fixed in the morning at sunrise each day and allowed to vary during the course of the day. Other species are treated as prognostic variables in the model. The model calculations are run through a sufficient number of diurnal cycles to achieve steady state. For all species of interest, 5 days of integration are sufficient in summer; for winter conditions, 30 days of integration are needed.

The starting value of the O_3 mixing ratio in the model calculations is set at 40 ppbv, which is representative of the planetary boundary layer (PBL). The CH_4 mixing ratio is 1600 ppbv. In the PBL, CO is scaled to NO_x according to concurrent measurement of the two species at the site (P. D. Goldan, private communication, 1986). The scaling gives about 250 ppbv of CO at NO_x level less than 0.5 ppbv and about 750 ppbv of CO at 10 ppbv of NO_x . The CO level at the low NO_x level is probably 10–40% too high judged by surface level measurements made near this latitude [e.g., Hoell et al., 1985; Pratt and Falconer, 1979; Junge et al., 1971]. However, it will be shown that the difference in CO level has little influence on the conclusions drawn from this study.

Since the photochemistry of O_3 is strongly affected by the concentrations of NMHC, it is important to define accurately the abundance of natural and anthropogenic hydrocarbons. In general, anthropogenic hydrocarbons are transported to the site from the Denver metropolitan area. Greenberg and Zimmerman [1984] measured most of the important anthropogenic hydrocarbons and reported their average mixing ratios. In our model calculations the mixing ratios of anthropogenic hydrocarbons are determined as follows. First, we assume that the concentrations of anthropogenic hydrocarbons are linearly proportional to the concentration of NO_x and their ratios are determined from the average values of hydrocarbons measured by Greenberg and Zimmerman [1984]. The summertime average NO_x is about 0.8 ppbv in the day [Williams et al., 1984]. The anthropogenic NMHC mixing ratios included in the model are 2.5, 1.5, 1.0, 0.5, and 0.2 ppbv for C_2H_6 , C_3H_8 , C_4H_{10} , C_2H_4 , and C_3H_6 , respectively. These values are set to be about 20% higher than the average values of these species observed at this site in the summer by Greenberg and Zimmerman [1984] in order to account for hydrocarbons that are measured by them but not included in the model.

The anthropogenic NMHC included in our model are probably slightly lower than the amount present at the site (P. R. Zimmerman, private communication, 1986) because NMHC with carbon number greater than 10 and oxygenated hydrocarbons were not measured. In addition, the relative abundance of anthropogenic NMHC will change with the age of air mass due to differing rates of photochemical reactivities. The mixing ratios of highly reactive species should decrease faster than less reactive species. However, the linear scaling of all anthropogenic NMHC with NO_x does not allow for the differentiation between NMHC with different lifetimes. This tends to underestimate the reactivity of NMHC at high NO_x and to overestimate it at low NO_x . However, since we are not trying to simulate a specific event and there are substantial uncertainties in the photochemistry of NMHC, we believe that

this representation of the anthropogenic NMHC and their photochemistry is adequate.

The average concentrations of the natural hydrocarbons at about a height of 1 m measured at Niwot Ridge in the summer were 0.63 ppbv for isoprene and about 0.35 ppbv for the terpenes [Greenberg and Zimmerman, 1984]. If these values were characteristic of the total PBL, they would have a very large impact on the photochemistry of O_3 and odd hydrogen species. However, a PBL model simulation [Hov et al., 1983] of the vertical distributions of terpenes shows that under normal summer atmospheric conditions the mixing ratios of terpenes decrease sharply with height in the first 20 m of the surface air. This is because the vertical turbulent mixing is inefficient near the surface where the hydrocarbons are emitted and they are rapidly destroyed photochemically before they have an opportunity to mix throughout the PBL. Hov et al. [1983] calculated average mixing ratios of terpenes in the PBL that are more than a factor of 5 lower than the surface values. We have made a similar calculation for isoprene and found a similar decrease of mixing ratio with height (M. Trainer et al., Impact of natural hydrocarbons on hydroxyl and peroxy radicals at a remote site, submitted to *Journal of Geophysical Research*, 1987). Therefore the average mixing ratio for isoprene and terpenes in the PBL should be about 0.1 and 0.05 ppbv, respectively. Since the photochemistry of terpenes is poorly known, we assume that all natural hydrocarbons are in the form of isoprene with a mixing ratio of 0.15 ppbv in the PBL and negligible above. At this level the natural hydrocarbons will increase the photochemical production of O_3 by about 20%, a significant amount but well within the uncertainty of our model.

The HNO_3 concentration is scaled to NO_x , $[\text{HNO}_3] \sim 0.3 [\text{NO}_x]$. Because there is less HNO_3 than NO_x , the conversion of HNO_3 to NO_x is negligible. Thus the conversion of NO_x to HNO_3 constitutes a real sink for NO_x . Finally, NO_x and the anthropogenic hydrocarbons are assumed to be well mixed in the PBL.

Solar insolation for July 21 conditions is assumed to represent the average summer value and January 21 insolation for the average winter value. The overhead O_3 column density is fixed at 313 Dobson units in the summer and 333 Dobson units in the winter [Dütsch et al., 1970]. The ground albedo is set at 10%. The H_2O level is fixed at 1% in the summer and 0.33% in the winter. The temperature changes with local time as prescribed by observed mean values. Values of photolysis rate at noontime are listed in the end of Table 1.

The surface deposition of trace gases in the PBL is included in the model by adding a sink term that is equal to the surface deposition velocity divided by the thickness of the PBL. For ozone the choice of deposition velocity is of fundamental importance, since the lifetime of tropospheric ozone can depend on the rate that ozone is destroyed at the surface, especially in the winter. During the summer an ozone surface deposition velocity of 0.5 cm s^{-1} [Aldaz, 1969; Galbally and Roy, 1980; Wesely et al., 1981; Lenschow et al., 1982; Colbeck and Harrison, 1985] is used. The data are sparse on O_3 deposition in the winter. For snow the surface resistance to O_3 uptake is large. A value of 11 s cm^{-1} was observed by Colbeck and Harrison [1985]. Galbally and Roy [1980] reported a median value of 16 s cm^{-1} with a great deal of variation, while Wesely et al. [1981] reported a value of about 34 s cm^{-1} with small variation. Wesely [1983] estimated from their experiments that

TABLE 1a. Reaction Rate Constants

Reaction	Rate Constant
$O(^1D) + H_2O \rightarrow 2OH$	2.2×10^{-10}
$O(^1D) + CH_4 \xrightarrow{O_2} CH_3O_2 + OH$	1.4×10^{-10}
$OH + CH_4 \xrightarrow{O_2} CH_3O_2 + H_2O$	$2.4 \times 10^{-12} \exp(-1710/T)$
$O(^1D) + H_2 \xrightarrow{O_2} HO_2 + OH$	1.0×10^{-10}
$OH + H_2 \xrightarrow{O_2} HO_2 + H_2O$	$6.1 \times 10^{-12} \exp(-2030/T)$
$OH + CO \rightarrow HO_2 + CO_2$	$1.5 \times 10^{-13} (1 + 0.6 p(\text{atm})) 10^{-11}$
$OH + HO_2 \rightarrow H_2O + O_2$	$(7 + 4 p(\text{atm})) 10^{-11}$
$OH + O_3 \rightarrow HO_2 + O_2$	$1.6 \times 10^{-12} \exp(-940/T)$
$HO_2 + O_3 \rightarrow OH + 2O_2$	$1.4 \times 10^{-14} \exp(-580/T)$
$HO_2 + HO_2 \rightarrow H_2O_2 + O_2$	$[1.9 \times 10^{-33} M \exp(980/T) + 2.2 \times 10^{-13} \exp(620/T)]$ $(1 + 1.4 \times 10^{-21} \exp(2200/T) H_2O)^*$
$OH + H_2O_2 \rightarrow HO_2 + H_2O$	$3.1 \times 10^{-12} \exp(-187/T)$
$HO_2 + NO \rightarrow NO_2 + OH$	$3.7 \times 10^{-12} \exp(240/T)$
$NO + O_3 \rightarrow NO_2 + O_2$	$1.8 \times 10^{-12} \exp(-1370/T)$
$OH + HNO_3 \rightarrow H_2O + NO_3$	$9.4 \times 10^{-15} \exp(778/T)$
$NO_3 + NO \rightarrow 2NO_2$	2×10^{-11}
$NO_2 + O_3 \rightarrow NO_3 + O_2$	$1.2 \times 10^{-13} \exp(-2450/T)$
$CH_3O_2 + HO_2 \rightarrow CH_3OOH + O_2$	$7.7 \times 10^{-14} \exp(1300/T)$
$CH_3O_2 + CH_3O_2 \rightarrow 2HO_2 + 2CH_2O$ (a)	$K = 1.6 \times 10^{-13} \exp(220/T)$
$\quad \quad \quad \rightarrow CH_2O + CH_3OH$ (b)	$K_a = 0.38 \text{ K}, K_b = 0.62 \text{ K}$
$CH_3OOH + OH \rightarrow CH_3O_2 + H_2O$ (a)	$K = 1 \times 10^{-11}$
$\quad \quad \quad \rightarrow CH_2O + OH + H_2O$ (b)	$K_a = 0.56 \text{ K}, K_b = 0.44 \text{ K}$
$CH_3O_2 + NO \xrightarrow{O_2} HO_2 + CH_2O + NO_2$	$4.2 \times 10^{-12} \exp(180/T)$
$CH_2O + OH \xrightarrow{O_2} HO_2 + H_2O + CO$	1×10^{-11}
$OH + NO \rightarrow HNO_2$	2×10^{-12}
$NO + NO_2 + H_2O \rightarrow 2HNO_2$	6×10^{-37}
$O(^1D) + M \rightarrow O(^3P)$	2.88×10^{-11}
$OH + HO_2NO_2 \xrightarrow{O_2} \text{products}$	$1.3 \times 10^{-12} \exp(380/T)$
$OH + C_2H_6 \xrightarrow{O_2} C_2H_5O_2$	$1.86 \times 10^{-11} \exp(-1231/T)$
$C_2H_5O_2 + NO \xrightarrow{O_2} CH_3CHO + NO_2 + HO_2$	$3.7 \times 10^{-12} \exp(240/T)$
$OH + C_3H_8 \xrightarrow{O_2} C_3H_7O_2$	$1.2 \times 10^{-11} \exp(-679/T)$
$C_3H_7O_2 + NO \xrightarrow{O_2} CH_3COCH_3 + NO_2 + HO_2$	$3.7 \times 10^{-12} \exp(240/T)$
$OH + C_2H_4 \xrightarrow{O_2} C_2H_4OHO_2$	$2.18 \times 10^{-12} \exp(387/T)$
$C_2H_4OHO_2 + NO \xrightarrow{O_2} 2CH_2O + NO_2 + HO_2$	$3.7 \times 10^{-12} \exp(240/T)$
$OH + C_3H_6 \xrightarrow{O_2} C_3H_6OHO_2$	$4.1 \times 10^{-12} \exp(544/T)$
$C_3H_6OHO_2 + NO \rightarrow CH_2O + CH_3CHO + NO_2 + HO_2$	$3.7 \times 10^{-12} \exp(240/T)$
$O_3 + C_2H_4 \rightarrow CH_2O + 0.4CH_2O_2 + 0.4CO + 0.1HO_2$	$2.57 \times 10^{-14} \exp(-2828/T)$
$O_3 + C_3H_6 \rightarrow 0.5CH_2O + 0.5CH_3CHO + 0.2CH_2O_2 + 0.2CH_3CHO_2 + 0.3CO + 0.2HO_2 + 0.1OH + 0.2CH_3O_2$	$7 \times 10^{-15} \exp(-1900/T)$
$CH_2O_2 + NO \rightarrow NO_2 + CH_2O$	7×10^{-12}
$CH_2O_2 + NO_2 \rightarrow NO_3 + CH_2O$	7×10^{-13}
$CH_2O_2 + SO_2 \rightarrow SO_4^{2-} + CH_2O$	6.7×10^{-14}
$CH_2O_2 + H_2O \rightarrow \text{products}$	3.3×10^{-18}
$CH_3CHO_2 + NO \rightarrow NO_2 + CH_3CHO$	7×10^{-12}
$CH_3CHO_2 + NO_2 \rightarrow NO_3 + CH_3CHO$	7×10^{-13}
$CH_3CHO_2 + SO_2 \rightarrow SO_4^{2-} + CH_3CHO$	6.7×10^{-14}
$CH_3CHO_2 + H_2O \rightarrow \text{products}$	3.3×10^{-18}
$OH + CH_3CHO \xrightarrow{O_2} CH_3COO_2 + H_2O$	$6.7 \times 10^{-12} \exp(250/T)$
$CH_3COO_2 + NO_2 \rightarrow PAN$	4.77×10^{-12}
$PAN \rightarrow CH_3COO_2 + NO_2$	$2 \times 10^{-16} \exp(-13543/T)$
$CH_3COO_2 + NO \xrightarrow{O_2} CH_3O_2 + NO_2 + CO_2$	$3.7 \times 10^{-12} \exp(240/T)$
$OH + C_4H_{10} \xrightarrow{O_2} C_4H_9O_2$	$1.76 \times 10^{-11} \exp(-558/T)$
$C_4H_9O_2 + NO \rightarrow 0.9NO_2 + 0.9HO_2 + 0.6CH_3CHO + 0.1C_2H_5CHO + 0.5CH_3COC_2H_5 + 0.1 \text{ nitrate}$	$3.7 \times 10^{-12} \exp(240/T)$

TABLE 1a. (continued)

Reaction	Rate Constant
$\text{OH} + \text{C}_2\text{H}_5\text{CHO} \xrightarrow{\text{O}_2} \text{C}_2\text{H}_5\text{COO}_2 + \text{H}_2\text{O}$	$2. \times 10^{-11}$
$\text{C}_2\text{H}_5\text{COO}_2 + \text{NO}_2 \rightarrow \text{PPN}$	4.77×10^{-12}
$\text{PPN} \rightarrow \text{C}_2\text{H}_5\text{COO}_2 + \text{NO}_2$	$2. \times 10^{16} \exp(-13543/T)$
$\text{C}_2\text{H}_5\text{COO}_2 + \text{NO} \xrightarrow{\text{O}_2} \text{C}_2\text{H}_5\text{O}_2 + \text{NO}_2 + \text{CO}_2$	$3.7 \times 10^{-12} \exp(240/T)$
$\text{OH} + \text{CH}_3\text{COC}_2\text{H}_5 \xrightarrow{\text{O}_2} \text{C}_2\text{H}_4\text{O}_2\text{COCH}_3 + \text{H}_2\text{O}$	$1 \times 10^{-11} \exp(-330/T)$
$\text{C}_2\text{H}_4\text{O}_2\text{COCH}_3 + \text{NO} \xrightarrow{\text{O}_2} \text{NO}_2 + \text{CH}_3\text{CHO} + \text{CH}_3\text{COO}_2$	$3.7 \times 10^{-12} \exp(240/T)$
$\text{N}_2\text{O}_5 \rightarrow \text{NO}_2 + \text{NO}_3$	$6.81 \times 10^{-6} \exp(-9884/T)$ $(1 + \text{M} \times 4 \times 10^{-20} \exp(951/T)) \text{ M}$
$\text{NO}_3 + \text{NO}_2 \xrightleftharpoons[\text{M}]{\text{M}} \text{N}_2\text{O}_5$	$K_{\text{eq}} = 1.2 \times 10^{-27} \exp(11180/T)$
$\text{HO}_2 + \text{NO}_2 \xrightleftharpoons[\text{M}]{\text{M}} \text{HO}_2\text{NO}_2$	$K_{\text{eq}} = 2.33 \times 10^{-27} \exp(10870/T)$

Units are $\text{cm}^6 \text{s}^{-1}$ for termolecular reaction, $\text{cm}^3 \text{s}^{-1}$ for bimolecular reaction, and s^{-1} for unimolecular reaction.

*Kircher and Sander [1984].

the surface resistance for agricultural land, rangeland, and nonforested wetland with snow to be about 30 s cm^{-1} , i.e., a deposition velocity of less than 0.03 cm s^{-1} . He also estimated that the surface resistance to O_3 for forested areas in cold weather is about 20 s cm^{-1} for near-neutral and nocturnal cases and about 3 s cm^{-1} for daytime conditions. Based on these measurements, we assume a daytime averaged O_3 deposition velocity over continental areas in the winter to be 0.1 cm s^{-1} . At night the deposition velocity of O_3 and other species is assumed to be negligible because the formation of a nocturnal inversion layer prevents efficient mixing to the surface.

The deposition velocity for NO_2 measured over various surfaces under summer conditions ranges from 0.3 to 0.8 cm s^{-1} , while the value for NO is much lower [Rogers *et al.*, 1977; Judeikis and Wren, 1978; Bottger *et al.*, 1978; Wesely *et al.*, 1982]. We assume a daytime value of 0.4 cm s^{-1} for NO_x in our model. There are few data on the deposition velocity of NO_x in winter conditions. This will be discussed in more detail later.

For other species that can be significantly removed from the atmosphere by surface deposition, no seasonal adjustments are made. A daytime deposition velocity of 1 cm s^{-1} is adopted for HNO_3 [Huebert and Robert, 1985]. In the absence of published results we arbitrarily assumed daytime deposition velocities for several key secondary reaction products, 0.5 cm s^{-1} for H_2O_2 and CH_3OOH and 0.1 cm s^{-1} for the aldehydes and ketones. These deposition velocities are low enough that

they do not have significant impact on the outcomes of the model.

Although explicit transport is neglected in the model calculations, the dilution effect of trace gases in the PBL due to the rise of the top of the PBL (i.e., the inversion height) during daytime in summer is included as follows. The rise of the height of the top of the PBL in the day used in the calculation is identical to that described by Kaimal *et al.* [1976]. Above the PBL, the mixing ratios of trace gases are assumed to be those of clean continental air: $40 \text{ ppbv } \text{O}_3$, $0.01 \text{ ppbv } \text{NO}_x$, $200 \text{ ppbv } \text{CO}$, and $1600 \text{ ppbv } \text{CH}_4$. The mixing ratios of NMHC and their secondary products such as aldehydes and ketones, unless noted otherwise, are assumed to be negligibly small above the PBL compared to those in the PBL. Thus, when the top of the PBL rises in the day, trace gases in the PBL are diluted by the clean air above the PBL. Horizontal dilution is not included in the calculation, but its effect will be discussed later. Dilution effects are not included in the model calculations for winter conditions.

COMPARISON OF CALCULATION AND MEASUREMENT

In Figure 2 the calculated and measured values of Q are shown for summer conditions. The measured values are the afternoon O_3 values from Figure 1a minus the morning values. The calculated values represent two cases of the model.

TABLE 1c. Photolysis Rates

Reaction	Rate
(R1) $\text{O}_3 + h\nu \rightarrow \text{O}(^1D) + \text{O}_2$	2.95×10^{-5}
(R2) $\text{NO}_2 + h\nu \rightarrow \text{NO} + \text{O}$	9.30×10^{-3}
(R3) $\text{H}_2\text{O}_2 + h\nu \rightarrow 2 \times \text{OH}$	1.07×10^{-5}
(R4) $\text{HNO}_3 + h\nu \rightarrow \text{OH} + \text{NO}_2$	8.4×10^{-7}
(R5a) $\text{CH}_2\text{O} + h\nu \rightarrow 2 \times \text{HO}_2 + \text{CO}$	2.53×10^{-5}
(R5b) $\text{CH}_2\text{O} + h\nu \rightarrow \text{H}_2 + \text{O}$	5.56×10^{-5}
(R6) $\text{NO}_3 + h\nu \rightarrow \text{NO}_2 + \text{O}$	7.32×10^{-2}
(R7) $\text{N}_2\text{O}_5 + h\nu \rightarrow \text{NO}_2 + \text{NO}_3$	3.35×10^{-5}
(R8) $\text{HNO}_2 + h\nu \rightarrow \text{OH} + \text{NO}$	1.86×10^{-3}
(R9) $\text{HO}_2\text{NO}_2 + h\nu \rightarrow \text{HO}_2 + \text{NO}_2$	8.0×10^{-5}
(R10) $\text{CH}_3\text{CHO} + h\nu \rightarrow \text{CH}_3\text{O}_2 + \text{HO}_2 + \text{CO}$	3.7×10^{-6}
(R11) $\text{CH}_3\text{OOH} + h\nu \rightarrow \text{CH}_2\text{O} + \text{OH} + \text{HO}_2$	$j_{11} = 0.7 \times j_3$
(R12) $\text{RCHO} + h\nu \rightarrow \text{C}_2\text{H}_5\text{O}_2 + \text{CO} + \text{HO}_2$	$j_{12} = j_{10}$
(R13) $\text{MEK} + h\nu \rightarrow \text{CH}_3\text{CO}_3 + \text{C}_2\text{H}_5\text{O}_2$	$j_{13} = j_{5a}$

Units are s^{-1} . Calculated for clear sky conditions, zenith angle = 21° , column $\text{O}_3 = 313$ Dobson units, surface at 3 km , and albedo = 0.1 .

TABLE 1b. Termolecular Reactions

Reaction	Rate Constant
$\text{OH} + \text{NO}_2 + \text{M} \rightarrow \text{HNO}_3$	$K_0^{300} = 2.6 \times 10^{-30}$, $n = 3.2$ $K_\infty^{300} = 2.4 \times 10^{-11}$, $m = 1.3$
$\text{HO}_2 + \text{NO}_2 + \text{M} \rightarrow \text{HO}_2\text{NO}_2$	$K_0^{300} = 2.3 \times 10^{-31}$, $n = 4.6$ $K_\infty^{300} = 4.2 \times 10^{-12}$, $m = 0$.

$$K = \left(\frac{K_0(T)}{1 + K_0(T)[\text{M}]/K_\infty(T)} \right) 0.6^{1.1 + [\log_{10}(K_0(T)[\text{M}]/K_\infty(T))]^2 - 1}$$

$$K_0(T) = K_0^{300}(T/300)^{-n}$$

$$K_\infty(T) = K_\infty^{300}(T/300)^{-m}$$

Units are $\text{cm}^6 \text{s}^{-1}$. $[\text{M}]$ air density (molecules/ cm^3).

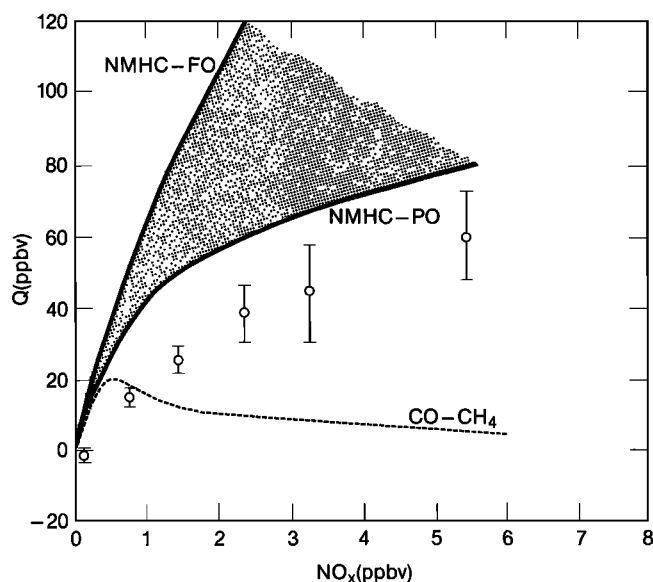
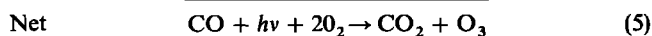
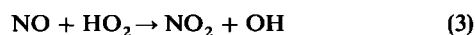


Fig. 2. Model-calculated daytime change in ozone (from sunrise to 1630 MST) for the summer clear sky conditions is compared to the observed difference between the afternoon (1400–1900 MST) and morning (0700–1100 MST) for clear sky conditions. The dashed line is calculated from a model without NMHC. The shaded area represents calculated values from a model with anthropogenic NMHC. The lower envelope of the shaded area is calculated by assuming no overnight retention of secondary hydrocarbons (NMHC-PO), while the upper envelope assumes buildup of secondary hydrocarbons to their steady state values (NMHC-FO).

The dashed line, which we will refer to as the CO-CH₄ case, is the model calculation without NMHC. The shaded area bounded by the solid lines represents the possible range of Q obtained by including the effects of NMHC in the model calculation. The lower solid line represents the model calculation where the diurnal change in the planetary boundary layer dilutes the secondary hydrocarbon products, e.g., aldehydes and ketones. Thus these compounds do not accumulate sufficiently to influence the photochemistry. This limit will be referred to as the NMHC partial oxidation (NMHC-PO) case. The upper solid line is calculated assuming that the trace gases are not diluted by the change of the height of PBL. An example for this may be found in a stagnant anticyclonic system where the trace gases in the afternoon PBL are not dispersed during the night when the nocturnal inversion layer is formed. These trace gases will be mixed down the following morning after the inversion layer breaks down. In this case the secondary hydrocarbons can accumulate over several days. Here the secondary hydrocarbon products accumulate to their steady state values in 2–3 days, thus exerting their maximum influence on O₃ production. This limit will be referred to as the NMHC full oxidation (NMHC-FO) case.

In the CO-CH₄ case, ozone production is a by-product of the catalytic oxidation of CO and CH₄ by NO_x and odd hydrogen radicals. For CO this cycle is given by

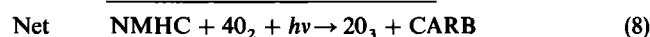
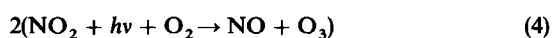
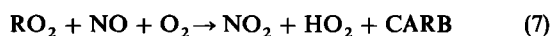


CH₄ can play a role similar to that of CO but at a smaller rate. In addition, depending on the ambient conditions (for

example, NO_x mixing ratio) and the detailed photochemical processes assumed, CH₄ can be a small source or sink for hydrogen radicals.

The estimated uncertainty in Q predicted by the model for the CO-CH₄ case is $\pm 50\%$. Considering both the uncertainty in the model prediction and the variability of the measurements for elevated NO_x levels, the CO-CH₄ case underpredicts the value of Q deduced from observations at the site. This discrepancy is caused by the reaction of NO₂ with OH to form nitric acid. At higher NO_x levels this process rapidly depletes the odd hydrogen radicals and strongly suppresses the photochemistry.

The inclusion of NMHC substantially alters the predicted Q at higher NO_x levels. This process can be represented by the simplified scheme



where R stands for hydrocarbon radical and CARB denotes carbonyl compounds. Reaction (8) shows that two O₃ molecules are produced for every NMHC oxidized. In addition, the carbonyl compounds may undergo further photochemical reactions which will result in a significant net gain of hydrogen radicals and, in turn, produce more O₃. The shaded area in Figure 2 can be interpreted as representing the uncertainty due to various levels of accumulation of carbonyl compounds in the PBL.

The sensitivity of the O₃ production P to uncertainties in the NMHC concentrations has been tested by changing these concentrations in the model. When NO_x is less than 4 ppbv, the sensitivity is relatively small; for example, a factor of 2 change in NMHC concentrations results in less than a 30% change in the O₃ production. The change increases to 50% at 6.5 ppbv of NO_x. The limitations implicit in the use of the simple relation to deduce the NMHC concentrations coupled with the lack of understanding of the photochemistry of NMHC are the two largest sources of uncertainty in this model. We estimate the uncertainty in the model-predicted Q , including the effects of NMHC, to be approximately a factor of 1.5 below 1 ppbv of NO_x, a factor of 2 for NO_x levels between 1 ppbv and 5 ppbv, and a factor of 3 for NO_x levels above 5 ppbv.

In comparing the model predictions with measurements one has to note that the measurement site at Niwot Ridge is significantly influenced by a single source of anthropogenic emissions. Since the transport time from this source to the site is less than 1 day, the secondary reaction products of hydrocarbon oxidation cannot accumulate in the sampled air masses. Consequently, the measured Q should be compared to values near the bottom of the shaded area of the model calculation, i.e., the NMHC-PO case. Although the model calculated values of Q lie above the measured values, the differences between predicted and measured Q values are well within the estimated uncertainty except for NO_x levels below 1 ppbv. Below 1 ppbv of NO_x, model calculations with or without NMHC overestimate the O₃ increase by a factor of 2. It is suspected that the model calculations are overestimating odd

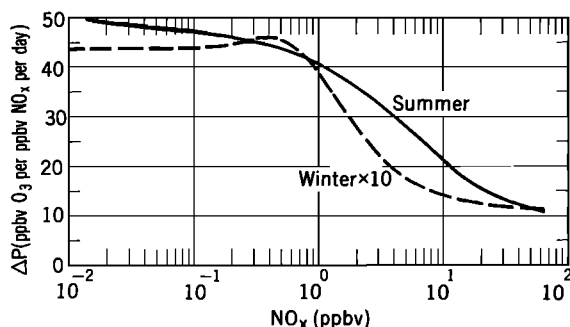


Fig. 3. The values of ozone production per unit NO_x per day, ΔP , from the NMHC-FO model are plotted as a function of NO_x mixing ratios. A constant NMHC to NO_x ratio is assumed; see text for detail. The solid line gives summer values. The dashed line gives the winter values multiplied by 10.

hydrogen radical concentrations. This has been recognized previously [Roberts *et al.*, 1984; Parrish *et al.*, 1986b], but the causes of this overestimation have not been established.

A clear feature that emerges from Figure 2 is the nonlinear character of Q as a function of NO_x level. This is evident in both the observed ozone mixing ratios and the calculations that include the NMHC. It is less pronounced for the calculation with the accumulation of secondary hydrocarbons. This is expected because the production of HO_2 and RO_2 radicals from secondary hydrocarbons compensates the increased loss of OH radicals due to their reaction with NO_2 at higher levels of NO_x .

Dilution due to horizontal transport will have a similar effect as vertical dilution. Namely, its major effect is to prevent the accumulation of secondary hydrocarbons.

In the winter data (Figure 1b) the afternoon average O_3 concentrations are slightly higher than the morning values, but the difference may not be statistically significant. This may be simply due to the rise of the inversion layer in the daytime and the mixing of upper level O_3 down into the PBL. The photochemical production and loss rates of O_3 are so small that the O_3 distribution is controlled by transport. Therefore quantitative comparison of the observed daytime O_3 change at this site with our simple model that does not incorporate realistic transport processes is not meaningful.

COMPARISON WITH OTHER OBSERVATIONS

It has been shown that the summer ozone observations at Niwot Ridge can be reasonably well matched by model-predicted diurnal ozone profiles that are chosen to simulate physical parameters and chemical species concentrations observed at Niwot Ridge. In this section the data and model calculations are compared to simultaneous O_3 - NO_x observations made in the summer at other sites in the United States.

At present, there are few published reports of simultaneous measurements of NO_x and O_3 at rural sites with sufficient amounts of data to estimate the value of Q . Kelly *et al.* [1984a] observed NO_x and O_3 at three sites located in South Dakota, Louisiana, and Virginia. The average mixing ratio of NO_x at these sites was 2.9 ppbv, and the inferred Q was 17 ppbv O_3 . At Niwot Ridge including all sky conditions, the O_3 increase corresponding to an NO_x level of 2.9 ppbv was approximately 25 ppbv, i.e., about 50% greater than the value observed by Kelly *et al.* [1984a]. This difference can be explained by the altitude difference of the sites. Our station is at

3 km altitude where the O_3 production efficiency is estimated to be about 40% higher than at sea level due to the larger photolysis rates of O_3 that lead to the production of $\text{O}(^1\text{D})$ and hence to OH radicals.

Research Triangle Institute [1975] made measurements of O_3 and its precursors at five rural stations in the summer of 1974. The mean NO_2 mixing ratios at these stations are similar to each other, ranging from 3 ppbv to about 5 ppbv in the afternoon. Assuming that NO is about 1/3 of NO_2 [Williams *et al.*, 1984], NO_x mixing ratios ranging from 4 to 7 ppbv are derived. The mean diurnal O_3 distributions reported have essentially the same shape as that at our station. With the exception of one station, McHenry, the daytime O_3 increase Q is about 47 ppbv. (The McHenry station in the state of Maryland has an elevation of 884 m above sea level. Its observed daily increase in O_3 is only about 20 ppbv, while the afternoon NO_x is relatively high at about 7 ppbv. Research Triangle Institute [1975] did not find any obvious cause for the low O_3 buildup at McHenry but noticed that the O_3 buildup had been 50% higher in the previous summer. For this reason, the data from the McHenry site are excluded from the present discussion.) This Q value after correction for elevation and cloud cover is compatible with the Q values inferred from the Kelly *et al.* [1984a] measurements and the value deduced for Niwot Ridge from Figure 1.

In comparing these data the limitation of present NO_x -measuring techniques should be recognized. We presently know that NO_2 to NO surface conversion techniques used in most chemiluminescence detectors can also convert organic nitrates, e.g., peroxyacetyl nitrate (PAN), and in some cases HNO_3 to NO [Kelly *et al.*, 1984b; Grosjean and Harrison, 1985; F. C. Fehsenfeld *et al.*, A ground-based intercomparison of NO, NO_x , and NO_y measurement techniques, submitted to *Journal of Geophysical Research*, 1987]. Accordingly, NO_x as measured by these instruments is an upper limit. Depending on the air masses sampled during the summer, the measurement can overestimate the NO_x concentration by a factor as large as 3 [Fahey *et al.*, 1986]. Thus the value of Q estimated above for the data of Kelly *et al.* [1984a] and Research Triangle Institute [1975] may actually correspond to lower NO_x levels.

NONLINEARITY IN OZONE PRODUCTION

One of the important observations that has been made concerning the net daily ozone change Q is the nonlinear relationship between Q and $[\text{NO}_x]$. Both, calculations and measurements, indicate that Q increases with NO_x more rapidly at low concentrations of NO_x . Since loss and transport of O_3 , L and T in equation (1), are almost independent of NO_x , the nonlinear dependence in Q is associated with the variation in photochemical production P with NO_x . This effect can be seen clearly in Figure 3, which shows a plot of the calculated

$$\frac{P}{[\text{NO}_x]} = \Delta P \quad (9)$$

versus $[\text{NO}_x]$. The quantity ΔP is the average daily ozone production per unit concentration of NO_x (i.e., ppbv O_3 per ppbv NO_x per day). The two curves in Figure 3 show ΔP for typical summer and winter conditions as calculated by the NMHC-FO model. The dependence of ΔP on the NO_x level is very similar for the two seasons with the summer values approximately a factor of 10 larger, reflecting the higher photo-

chemical activity in the summer. Because the shape of the seasonal curves for ΔP are similar, the following comments that are made for the summertime variation in ΔP with NO_x are also applicable to the wintertime case.

Our model calculation predicts, for NO_x levels below 500 pptv, that ΔP is independent of the NO_x level. The linear dependence of Q on NO_x at low levels of NO_x is observed in all model conditions that have been run (cf. Figure 2) and is also observed in summertime ozone measurements at Niwot Ridge. Between 0.5 and 5 ppbv of NO_x , however, ΔP decreases with increasing NO_x levels. According to the NMHC-FO calculation, ΔP decreases by a factor of 4 between 0.5 and 5 ppbv of NO_x (see Figure 3). Above 5 ppbv of NO_x , the NMHC-FO case indicates that ΔP becomes less dependent on NO_x . However, the other model cases show a sharper decline in ΔP for $[\text{NO}_x] > 1$ ppbv. The sharper decline is also observed in the measurements at Niwot Ridge (cf. Figure 2) and elsewhere [Research Triangle Institute, 1975]. However, at NO_x levels above 5 ppbv the continued rapid decrease in the observed ΔP may be due to the short residence time of NO_x and NMHC in the atmosphere. Under these conditions, NO_x and NMHC are not able to reach full O_3 -producing potential.

The decline in ΔP for $\text{NO}_x > 1$ ppbv is consistent with the findings of photochemical smog models [e.g., U.S. Environmental Protection Agency (EPA), 1977; Hov and Derwent, 1981; Costanza and Seinfeld, 1982; Sakamaki et al., 1982; Altshuler, 1986]. Those models are usually intended for ambient NO_x and NMHC levels substantially higher than the present study. Thus the ratio of NMHC to NO_x and the mixture of NMHC in the smog models are significantly different. According to our modeling study and the smog models the degree of non-linearity is a function of the ratio of NMHC to NO_x and the relative abundance of various NMHC.

The higher value of ΔP at lower NO_x suggests that the dilution of NO_x and NMHC by atmospheric turbulence and advection will enhance the efficiency of O_3 production. This phenomenon may have important implications for the global and regional tropospheric O_3 budgets. Previously, many one-dimensional, as well as two-dimensional, modeling studies have neglected the nonlinearity effect in evaluating the global budget of O_3 due to anthropogenic NO_x emissions [e.g., Liu, 1977; Fishman et al., 1979; Chameides and Tan, 1981; Crutzen and Gidel, 1983]. In these earlier studies, NO_x emissions were assumed to be dispersed over domains that are much greater than the real domain of emissions. Because of the nonlinearity in ozone production this approach results in a significant overestimation in the O_3 produced by anthropogenic NO_x emissions.

REGIONAL OZONE PRODUCTION

For a given region the ozone production could be obtained by integration of P . For a particular region of interest, however, the spatial and temporal distribution of the NO_x mixing ratio is not likely to be available. However, if the region is large enough, the NO_x emission into and removal from the atmosphere will occur primarily within its boundary, and O_3 production can be approximated by

$$S = E\tau \Delta P \quad (10)$$

where S is the total O_3 produced due to the NO_x emission E within the region, τ denotes the NO_x lifetime, and ΔP is the daily O_3 production per ppbv of NO_x .

Equation (10) can be viewed in two ways. First, the product $E\tau$ is equal to the total number of NO_x molecules within the region. Since ΔP is the O_3 production per unit NO_x per unit time, the final product is O_3 production per unit time. Alternatively the product, $\tau \Delta P$ can be rewritten,

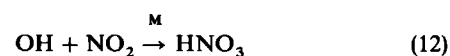
$$\tau \Delta P = \tau \left(\frac{P}{[\text{NO}_x]} \right) = \frac{P}{L[\text{NO}_x]} \quad (11)$$

where L is the rate of loss of NO_x . Thus $\tau \Delta P$ is equal to the number of O_3 molecules produced for each NO_x molecule destroyed. In steady state this equals the number of O_3 molecules produced for each NO_x molecule emitted. The total O_3 production S is obtained by multiplying $\tau \Delta P$ by the emission rate. In both approaches, τ and ΔP are assumed to be constant for each season in the region of interest over the lifetime of NO_x .

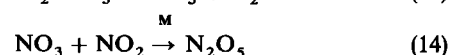
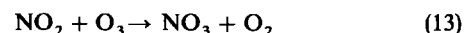
Even though equation (10) relates the O_3 production to NO_x lifetime, this is not intended to imply that NO_x alone is the rate limiting precursor of O_3 . Because in our calculations anthropogenic NMHC are assumed to be proportional to the NO_x concentration in our calculations, equation (10) contains NMHC implicitly. Profiles similar to those in Figure 3 and an equation equivalent to equation (10) could be generated for NMHC.

Equation (10) provides an important insight into evaluating the relative importance of CO and CH_4 versus NMHC in the production of O_3 from anthropogenic NO_x . Most O_3 is produced when the anthropogenic NO_x is within its first two lifetime periods after emission (in the summer 1–2 days [Chang et al., 1979]). Since the median rural NO_x level in the eastern United States is about 6.6 ppbv [Mueller and Hidy, 1983], our model shows that NMHC are essential in producing O_3 . Without NMHC the O_3 production would be reduced by a factor of 5. In the calculation with NMHC the O_3 production rate is essentially independent of the amount of CO present, implying a very small contribution for CO and CH_4 . In fact, the only way that substantial O_3 can be produced from the interaction of CO and CH_4 with anthropogenic NO_x is for a substantial amount of NO_x to be transported to the remote troposphere before it is removed from the atmosphere. PAN, which is a product of NMHC reactions and serves as a temporary reservoir and carrier for NO_x [Singh et al., 1985], can act as an agent to export anthropogenic NO_x to the remote troposphere. It is clear that the production of O_3 from the interaction of CO and CH_4 with anthropogenic NO_x depends critically on photochemistry and transport and is closely connected with NMHC. Previous estimates of the O_3 production that neglect the effect of NMHC are probably incorrect.

For the present approximation we assume that the NO_x lifetime τ in summer is determined during the day by the reaction of NO_2 with OH followed by rapid deposition of HNO_3 :



and at night by the reactions



followed by the conversion of NO_3 and N_2O_5 to HNO_3 on aerosols [Noxon, 1983; Platt et al., 1984] or in the gas phase

TABLE 2. Comparison of O₃ Production Parameters for Winter and Summer Conditions as Calculated From the NMHC-FO Model at Sea Level at 40°N Latitude With Clear Sky Conditions*

NO _x , ppbv	Season	ΔP ,* [O ₃]/[NO _x]	τ (NO _x), days	$\Delta P \times \tau$ (NO _x), [O ₃]/[NO _x]	[OH], cm ⁻³	τ (O ₃), days
0.1	summer	47	1.2	58.3	1.0E6	12
	winter	4.6	13	59.5	0.8E5	180
0.65	summer	43	0.6	25.7	2.0E6	9
	winter	5	4.5	22.7	1.9E5	102
1.5	summer	39	0.42	16.6	2.5E6	7
	winter	4.1	3.7	15	2.0E5	68
4	summer	29	0.4	11.6	2.6E6	6
	winter	2.7	4	10.5	1.7E5	39
10	summer	20	0.45	9.1	2.2E6	4.5
	winter	1.9	4.3	8.4	1.5E5	21

Read 1.0E6 as 1.0×10^6 . All values are diurnally averaged.

*O₃ molecules produced per NO_x molecule per day.

[Morris and Niki, 1974; Noxon, 1983; Atkinson et al., 1984; Platt et al., 1984]. The nighttime sinks for NO₂ are somewhat uncertain due to our lack of understanding of the details of the conversion mechanisms for NO₃ and N₂O₅ to HNO₃. The upper limit of these nighttime sinks is the total removal of N₂O₅, which is equal to twice the rate of reaction (13), removing two NO_x molecules at a time. Because of the difficulty in the quantitative treatment of the nighttime sink, in the following discussions the nighttime sink will be neglected unless noted otherwise.

The calculated lifetime of NO_x in the summer is consequently determined primarily by reaction (12) and thus depends on the OH concentration. The OH concentration, in turn, is determined by the mixing ratios of NO_x, NMHC, water vapor, and CO. Table 2 gives a list of the OH concentrations and NO_x lifetimes calculated by the model for summer and winter seasons as a function of NO_x mixing ratio at sea level for clear sky conditions. It is well known that the OH concentration and τ depend directly on the solar UV intensity and thus on season and/or cloud cover. However, the dependence of ΔP on solar UV intensity is equal in magnitude but opposite in sign to that of τ . Hence the product $\Delta P\tau$ and therefore S are essentially independent of season. Likewise, S is independent of cloud cover. In the same way, S deduced from NMHC-FO model has nearly the same value as the one deduced from the NMHC-PO model. For example, the calculated ΔP at 10 ppbv NO_x in the summer is 22 ppbv O₃ per ppbv NO_x per day for the NMHC-FO case and 12.5 for the NMHC-PO case, while the value of τ is 0.45 day for the former and 0.72 day for the latter. Thus, although both ΔP and τ are each subject to uncertainties of the order of a factor of 3 depending on the atmospheric chemical composition and the uncertainties in the attendant odd hydrogen radical chemistry, because of the conjugate relationship between ΔP and τ , the uncertainty in S is no larger than a factor of 2.

As discussed above, equation (10) may also be written for NMHC if they are the rate-limiting precursor for O₃. It can be shown that the seasonal invariance of O₃ production suggested by Table 2 will not change using this approach. For simplicity of discussion, let us assume that a surrogate hydrocarbon can be used to represent all the NMHC. Then the total O₃ produced, S , would be equal to the product of the emission rate of this hydrocarbon, its lifetime, and the daily O₃ production rate per ppbv of the hydrocarbon. Since the major

sink of the hydrocarbon would probably be the reaction with OH, the seasonal variation of S , assuming hydrocarbons to be the independent variable, would be the same as that shown in Table 2.

On the other hand, from the change of the product $\Delta P\tau$ with NO_x levels shown in Table 2, we note that both ΔP and τ decrease with increasing NO_x. The value of this product as a function of NO_x level is plotted in Figure 4. This enhances the nonlinear effect described previously and increases the uncertainty in our simplistic evaluation of the O₃ production. In the following discussion, regional ozone production will be derived using an approximate value of $\tau \Delta P$ from Figure 4 compatible with the assumed regional NO_x distribution. This provides a useful qualitative estimate for regional ozone production. Models that incorporate realistic transport and photochemical processes are needed to evaluate this production accurately.

OZONE PRODUCTION IN THE UNITED STATES IN SUMMER

In this section, equation (10) will be used to estimate ozone production for the United States in the summer season from anthropogenic and natural NO_x sources. In order to calculate S for anthropogenic and natural NO_x emission we choose an average NO_x level of the United States that is appropriate to each NO_x emission and then choose values for $\tau \Delta P$ corresponding to each level (cf. Figure 4). The NO_x levels in the United States can largely be attributed to anthropogenic sources. A majority of the anthropogenic NO_x is emitted in the eastern United States. In this region the median rural NO_x level is observed to be about 6.6 ppbv [Martinez and Singh, 1979; Ferman et al., 1981; Shaw and Paur, 1983; Mueller and Hidy, 1983]. In other areas of the United States the median NO_x levels are lower. For these levels we assume that $\tau_a \Delta P_a = 10$ ozone molecules formed for each NO_x emitted.

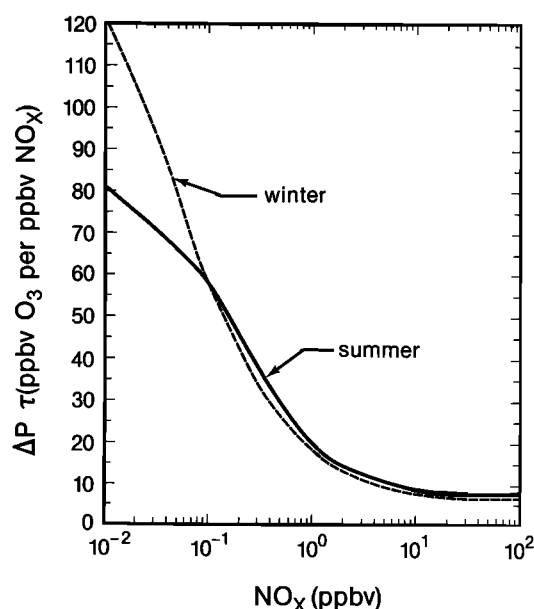


Fig. 4. The values of ozone molecules produced per NO_x molecule destroyed due to OH reaction with NO₂, $\Delta P\tau$, from the NMHC-FO model are plotted as a function of NO_x mixing ratios. The solid line gives summer values, and the dashed line gives winter values.

The anthropogenic emission rate E_a is 6×10^{12} g N yr⁻¹ in 1980 (Logan [1983], following U.S. EPA [1982]) with little seasonal variability. The subscripts "a" and "n" are used to denote values derived from anthropogenic and natural NO_x emissions, respectively.

Using the values derived above for E_a and $\tau \Delta P$, equation (10) yields

$$S_a = 5 \times 10^{13} \text{ g}$$

for O₃ produced from anthropogenic NO_x sources in the United States in the three summer months. Assuming the O₃ production to be uniform over the area of the United States for the summer months yields an average column O₃ production rate of $1 \times 10^{12} \text{ cm}^{-2} \text{ s}^{-1}$.

To estimate the production of ozone from natural sources, the NO_x levels attributable to natural NO_x emissions must be determined. This is equivalent to calculating S_n for the preindustrial era. Natural NO_x emissions are smaller than anthropogenic NO_x sources [Logan, 1983] and are more diffuse. Typical NO_x levels over the continental United States attributable to natural NO_x emissions would be 0.5 ppbv or less. The measurement of NO_x in rural and remote areas supports this limit [McFarland et al., 1979; Schiff et al., 1979; Kley et al., 1981; Helas and Warneck, 1981; Williams et al., 1984; Ridley et al., 1987]. For [NO_x] ≤ 0.5 ppbv we choose $\tau_n \Delta P_n = 32$ for ozone molecules produced per NO_x molecule emitted from natural sources. Thus, in the preindustrial United States natural NO_x was about 3 times as efficient in producing ozone as anthropogenic NO_x emission is at present.

NO_x has a variety of natural sources including soil emissions, lightning, and stratospheric subsidence [Logan, 1983]. Biogenic NO emissions from soils are estimated to range from 1×10^9 to $2 \times 10^{10} \text{ cm}^{-2} \text{ s}^{-1}$ in the summer, with average of about $3 \times 10^9 \text{ cm}^{-2} \text{ s}^{-1}$ [Galbally and Roy, 1978; Slemr and Seiler, 1984; Williams et al., 1985]. The average O₃ produced from this NO flux is estimated to be $1 \times 10^{11} \text{ cm}^{-2} \text{ s}^{-1}$. A range of 0.5×10^{11} to $2 \times 10^{11} \text{ cm}^{-2} \text{ s}^{-1}$ is obtained by adopting the uncertainty ranges of NO_x emissions given by Logan [1983]. NO_x production from lightning is estimated to be between 0.07×10^{12} and 0.7×10^{12} g N yr⁻¹ in the United States [Logan, 1983; Albritton et al., 1984]. Assuming that it is uniformly distributed and that roughly 50% of total emissions occur in summer [Turman and Edgar, 1982], a range of 7.6×10^{10} to $7.6 \times 10^{11} \text{ cm}^{-2} \text{ s}^{-1}$ is obtained for O₃ production. NO_x emissions from biomass burning in the United States are about 0.05×10^{12} to 0.15×10^{12} g N yr⁻¹, mostly from forest fires [Seiler and Crutzen, 1980; Logan, 1983]. Assuming even distribution and no seasonal variation, this would give a range of 2.7×10^{10} to $8.2 \times 10^{10} \text{ cm}^{-2} \text{ s}^{-1}$ of O₃ produced in the summer. When combined, the value of S_n deduced from these natural photochemical sources ranges from a low of about 1.5×10^{11} to as high as $10 \times 10^{11} \text{ cm}^{-2} \text{ s}^{-1}$, averaged over the United States in the summer. In addition, there is a significant addition of ozone to the troposphere from the stratosphere. The average cross-tropopause O₃ flux is estimated to be $5 \times 10^{10} \text{ cm}^{-2} \text{ s}^{-1}$ [Danielsen and Mohnen, 1977; Mahlman et al., 1980]. Therefore, in the summer the sum of the O₃ generated from natural NO_x and the direct O₃ flux from the stratosphere on average is substantially smaller than the anthropogenic O₃ source in the United States.

In comparing natural ozone production with anthropogenic ozone production in the United States the simplified picture

presented here overlooks several factors that can potentially influence the results. First, it should be noted that the same CO mixing ratios were used to calculate S_n and S_a . However, CO mixing ratios were probably 50% lower over the United States in the preindustrial era. A 50% reduction in the CO mixing ratio would result in a 30% reduction in S_n . Second, the distribution of NO_x sources must be taken into account. For example, a substantial amount of the NO_x from lightning is generated in the upper troposphere where the effective NO_x lifetime may be significantly longer than the lifetime in the lower troposphere. This is due to reduced scavenging of HNO₃ and regeneration of NO_x from HNO₃ [Liu et al., 1980, 1983] at higher elevations. As a result, O₃ production due to NO_x from lightning and the stratosphere could be substantially greater than the presented estimate indicates.

Likewise, depending on the effects of dilution by transport and inhomogeneities in emissions, the ozone production from anthropogenic NO_x sources may vary substantially from region to region in the United States. The average value presented above would suggest that S_a is approximately twice S_n . However, in the central and eastern United States, with average NO_x of about 7 ppbv as discussed above, the O₃ doubling time is less than a half day in the boundary layer. In this case, the effect of transport is relatively small, and the increase of O₃ concentration due to anthropogenic emissions is probably greater than the ratio of the O₃ sources derived above. Thus in the central and eastern United States, human activities probably contribute at least 50–80% of the O₃ in the summer. This conclusion is consistent with the elevated O₃ levels observed over large areas in the central and eastern United States [Research Triangle Institute, 1975; Vukovich et al., 1977, 1985; Cleveland et al., 1977; Spicer et al., 1979; Wolff and Liroy, 1980; Fehsenfeld et al., 1983; Kelly et al., 1984a]. A similar situation appears to exist for western Europe [Cox et al., 1975; Guicherit and Van Dop, 1977; Hov, 1984].

OZONE PRODUCTION IN THE NORTHERN HEMISPHERE

In the preceding section the summer ozone production in the United States associated with natural and anthropogenic NO_x emissions was estimated. The estimation of O₃ production on a regional level is satisfactory for the summer when the NO_x lifetime is short. In this case, ozone production in and near the region is largely associated with NO_x emitted within the region. This approach is not adequate for the United States in the winter. In winter, NO_x emitted in the United States can, during its lifetime (cf. Table 2), be transported well beyond the boundaries of the United States. In this case, equation (10) can still be used to estimate ozone formation but over a significantly larger area. In this section the ozone production associated with anthropogenic NO_x emissions will be compared to that due to natural emissions for the northern hemisphere (NH).

To do that, the model estimates for ΔP obtained from this study which are based on measurements made in rural locations in the United States are extrapolated to deduce ozone production in other areas of the world. Since the ambient conditions in such areas may have a substantially different mix of NMHC and NO_x in comparison with the rural United States, the ΔP calculated for these areas may be inaccurate. Of particular concern are estimates of ΔP for the forested regions of the tropics and subtropics where natural sources dominate the emissions of NMHC [cf. Greenberg and Zimmerman, 1984]

and NO_x [cf. *Turman and Edgar*, 1982; *Slemr and Seiler*, 1984; *Galbally and Roy*, 1978]. Even in the mid-latitudes, composition and/or levels may be quite different from that used to deduce ΔP above. On the other hand, the use of these results to estimate ozone production in relatively clean oceanic areas should be reasonably accurate since the observed concentrations of NMHC are small [*Rudolph and Ehhalt*, 1981; *Eichmann et al.*, 1979, 1980], as assumed in the model. The extrapolations in the following may be speculative; however, they provide a perspective on the global ozone budget that would otherwise be unavailable.

Logan [1983] estimated the global budget of NO_x . The four largest sources are fossil fuel combustion, biomass burning, lightning, and biogenic emissions with global source strengths of 21 (14–28), 12 (4–24), 8 (2–12), and 8 (4–16) in units of $10^{12} \text{ g N yr}^{-1}$, respectively, with the numbers in parentheses indicating the uncertainty. These sources are essentially land sources and mostly occur near the surface. Other sources in the lower troposphere are insignificant by comparison. Biomass burning is mostly of anthropogenic origin [*Seiler and Crutzen*, 1980]. Therefore, globally, the emissions from anthropogenic sources are probably more than twice as large as the natural emissions.

Seasonal variations of natural NO_x sources are quite different from the anthropogenic sources. *Turman and Edgar* [1982] reported the seasonal variation of the lightning trigger occurrence at dawn and dusk. In the NH, about 40% of the lightning triggers occur in the summer versus only about 5% in the winter. Biogenic NO_x emission from soils also peak strongly in the summer, as observations [*Slemr and Seiler*, 1983; *Williams et al.*, 1985] show a strong dependence of the NO_x emission rate on the soil temperature. On the other hand, the principal anthropogenic NO_x source in the United States, combustion, is essentially independent of season [*U.S. EPA*, 1982]. The combustion source for the rest of the world is probably slightly higher in the winter because the need for space heating is not offset by use of air conditioning as in the United States. Most of the NO_x emissions from biomass burning take place in the tropics and mainly during the dry season [*Seiler and Crutzen*, 1980]. In the NH the dry season in the tropics usually occurs in the winter. Therefore it can be concluded that in the NH winter the anthropogenic sources by far dominate the NO_x emissions. We estimate that the ratio of the anthropogenic emissions to natural emissions is about 10 to 1 in the NH in the winter.

Assuming that the only significant anthropogenic NO_x emissions in summer are from combustion sources and that $\Delta P_n \tau_n = 10$, the resultant O_3 production S_a from anthropogenic sources would be $(1.8 \pm 0.6) \times 10^{14} \text{ g}$ for the three summer months. Since essentially all emissions occur in the NH, this corresponds to a NH average column O_3 production S_a of $(1.1 \pm 0.3) \times 10^{11} \text{ cm}^{-2} \text{ s}^{-1}$.

These numbers can be compared with the recent results of *Fishman et al.* [1985]. In that study a one-dimensional PBL model was used and predicted a value of S_a that is about 20% greater than our estimate. Considering the large uncertainties in these two different approaches, the agreement is surprisingly good.

In the summer the natural and anthropogenic NO_x emissions are about the same in the NH. However, as before, we take the higher O_3 production potential into account and by assuming $\Delta P_n \tau_n = 32$, S_n is computed to be $3 \times 10^{11} \text{ cm}^{-2}$

s^{-1} , which is about 3 times greater than the value computed for anthropogenic emissions. The additional natural O_3 source associated with the cross-tropopause flux in summer is small compared to the photochemical production.

Considering the NH as a whole, O_3 production in summer is probably dominated by the photochemical production from natural NO_x sources. However, as stated in the preceding section, since the O_3 lifetime in summer is relatively short, long-range transport of O_3 will be limited. In this context, it should be noted that the time for doubling O_3 due to photochemical production is an important characteristic time for comparison with the long-range transport time. Table 2 shows that the doubling time for O_3 is shorter than a day when the NO_x level is greater than 1 ppbv. As a result, the O_3 distribution tends to be controlled by regional sources, especially in the PBL. For example, O_3 distribution in the tropics and subtropics should be dominated by the natural photochemical O_3 sources and sinks, while the anthropogenic source controls mid- and high-latitude ozone levels.

Both model calculations and observations show a substantially lower daily O_3 increase in the winter compared to the summer. Table 2 lists the model calculated O_3 production rate ΔP averaged over a day at various NO_x levels for winter conditions compared to that of summer conditions. The O_3 production rate ΔP is about a factor of 10 lower in the winter compared to the summer (cf. Figure 3). The seasonal change of ΔP is almost entirely due to the change in the odd hydrogen radical concentrations, which is represented by the change in the OH density. The density of HO_2 changes by about the same ratio. If one considers only daytime chemistry, the photochemical lifetime of NO_x is inversely proportional to the OH density. In this case the product $\Delta P \tau$ is essentially independent of season. Figure 4 and Table 2 show that this is true for almost all levels of NO_x . *Fishman et al.* [1986] used a different approach to estimate the O_3 production in the eastern United States and arrived at a similar conclusion. The large NO_x lifetime in the winter predicted here implies that the NO_x distribution from a constant emission source, such as anthropogenic combustion, will lead to higher NO_x concentrations in the winter compared to the summer. A two-dimensional simulation of the NO_x distribution from combustion emissions [*Crutzen and Gidel*, 1983] estimated 2–20 times higher NO_x mixing ratio in most of the NH in January compared to July, supporting this conclusion.

The above statement does not apply to NO_x introduced in the upper troposphere because HNO_3 is removed relatively slowly from this region, as discussed earlier. However, recent model calculations by *Kasting and Singh* [1985] showed that in the winter the formation of PAN may reduce NO_x in the upper troposphere by a factor of 10, thus reducing the O_3 production there to an insignificant level. This leaves the stratospheric intrusion as the only significant natural O_3 source in the winter.

In the lower troposphere of the NH, the NO_x in the winter is essentially all due to anthropogenic emission. It follows that the O_3 production in the lower troposphere in the winter is dominated by the anthropogenic source. Therefore the average column O_3 production in the NH due to combustion should range from 0.8×10^{11} to $1.5 \times 10^{11} \text{ cm}^{-2} \text{ s}^{-1}$, i.e., the same as in the summer. Biomass burning could contribute a production rate as large as this if one assumes that half of the global NO_x emissions due to biomass burning occur in the

NH winter season. Therefore the anthropogenic source of O_3 could be 3–6 times the natural source of O_3 in the winter NH.

The seasonal invariability of the O_3 production depends critically on the seasonal variation of the lifetime of NO_x . So far we have assumed that the lifetime of NO_x is primarily determined by the daytime chemistry. We think this is justified under summer conditions because the nighttime sink and dry deposition of NO_x account for less than 50% of the NO_x sink. In the winter the nonphotochemical sinks, formation of HNO_3 at night (cf. equations (13) and (14)) and deposition of NO_2 , could be substantial. If NO_3 at night is assumed to be totally removed from the atmosphere in the winter, NO_x would have a lifetime of only about 2 days in the boundary layer and the O_3 production would decrease by a factor of about 3. The factor would be doubled if N_2O_5 instead of NO_3 is totally removed because for each N_2O_5 reaction two NO_x molecules are removed. To remove NO_3 or N_2O_5 effectively, the product of the NO_3 or N_2O_5 reactions would need to be a stable species that is readily removed from the atmosphere such as HNO_3 or particulate nitrate. One mechanism that may lead to this is the interaction of N_2O_5 or NO_3 with wet aerosols in humid conditions, as suggested by Platt *et al.* [1984]. At relative humidities less than 50% there has been no observational evidence suggesting that this occurs in the atmosphere. Kinetic studies [Morris and Niki, 1974; Atkinson *et al.*, 1984] showed that NO_3 reaction with aldehydes probably resulted in the production of HNO_3 . However, the major removal process for NO_3 or N_2O_5 is probably not due to the reaction with aldehydes [Noxon, 1983; Platt *et al.*, 1984]. Furthermore, the production rate of aldehydes is also strongly seasonally dependent, yielding slower removal of NO_3 in the winter.

It is clear that nighttime chemistry of NO_x may play a major role in reducing the O_3 production in the winter. However, our current knowledge on the NO_3 and N_2O_5 is not adequate for a quantitative assessment. In this context, it should be noted that even in the case of total removal of NO_3 or N_2O_5 , the anthropogenic source of O_3 would still be comparable to the stratospheric O_3 flux.

The surface deposition of NO_x may also significantly shorten the NO_x lifetime in the continental boundary layer in the winter. There have been little data on the deposition velocity of NO_x in winter conditions. However, Wesely *et al.* [1982] reported a large surface resistance at night in the summer over a soybean field that resulted in a NO_2 deposition velocity as low as 0.05 cm s^{-1} . The large surface resistance observed during the summer night was attributed by them to low biogenic activity at night which will certainly be true during winter. This suggests slow surface deposition for NO_2 under winter conditions. Preliminary results from field measurements of the NO_x deposition velocity in winter conditions indicate its value to be significantly less than 0.2 cm s^{-1} (D. H. Stedman, private communication, 1986). The deposition velocity of NO and NO_2 over water surfaces is negligibly small because of their low solubility [Lee and Schwartz, 1981]. Assuming an average NO_x deposition velocity of 0.1 cm s^{-1} and a 500-m PBL height in the winter, the lifetime due to surface deposition would be about 6 days. Since some NO_x will be transported above the PBL, the lifetime should be longer. Therefore surface deposition probably will not affect the NO_x lifetime appreciably. This is substantiated by the calculation of Crutzen and Gidel [1983] that assumed constant seasonal dep-

osition velocity and still predicted much higher NO_x in the winter than the summer.

Reduction of the winter O_3 production may also come from removal of secondary products of NMHC reactions, such as organic nitrates, aldehydes, and organic acids. The long NO_x lifetime and increased stability of the secondary products allow more time for their removal by processes such as heterogeneous scavenging or surface deposition.

In the above discussion, we have also neglected the effect of the Arctic winter which has attracted extensive attention [Rahn and McCaffrey, 1979; Heintzenberg *et al.*, 1981; Barrie *et al.*, 1981]. In the Arctic winter night, NO_x , hydrocarbons, and other pollutants may accumulate and give rise to enhanced photochemical production of O_3 and other pollutants in the spring [Isaksen *et al.*, 1985; Barrie and Hoff, 1985]. In fact, the Arctic effect can be considered to be an extreme case of the winter effect shown in Table 2 by extending the lifetime of NO_x and O_3 production over winter into spring. The net effect is that the Arctic plume will delay part of the winter O_3 production until the spring. Without a realistic model we can not accurately estimate the reduction of the winter O_3 production due to the Arctic plume.

It is clear that our evaluation of the winter anthropogenic O_3 source leads to an overestimate. The uncertainties discussed above do not allow us to quantify the overestimation. However, the anthropogenic source is so much greater than the natural source that the former would need to be reduced by a factor of more than 10 to alter our conclusions.

LIFETIME OF OZONE

The seasonal variation of the photochemical lifetime of O_3 at 40°N is given in Table 2. The calculations in Table 2 are for sea level under clear sky conditions. The cloud cover should increase O_3 lifetime in the boundary layer by about 30%. In addition, above the boundary layer the O_3 lifetime is substantially longer than the values in Table 2 because of lower H_2O mixing ratios. Our calculations show that at 500 mbar the O_3 lifetimes are about 50% larger than those shown in Table 2. In estimating the O_3 lifetimes the expression for O_3 and other odd oxygen species are grouped following the designation of Levy *et al.* [1985]. In this approach the lifetime of O_3 is equal to the sum of the concentrations of all odd oxygen species (O_3) divided by the photochemical loss of odd oxygen. This expression provides a good representation of the net O_3 photochemical production and destruction. For example, NO_2 is considered to be one of the odd oxygen species because to a large extent the photolysis of NO_2 balances the reaction of O_3 with NO and does not result in either production or loss of O_3 . Reactions such as HO_2 and RO_2 with NO are counted as production terms for O_3 .

The lifetime of O_3 is about a factor of 10 longer in the winter than in the summer. The long lifetime of O_3 in the winter implies that O_3 will be transported over long distances. Once anthropogenically produced O_3 is transported to the relatively clean troposphere, the photochemical lifetime at mid-latitudes in the winter will be greater than 200 days. This is certainly longer than the characteristic time of zonal transport which is of the order of 30 days [Oort, 1983] and probably longer than the time of transport between mid-latitudes and lower latitudes in the NH. The latter transport time is difficult to estimate but is probably less than 3 months.

The photochemical lifetime of O_3 in the winter in mid-

latitudes is so long that the O_3 lifetime is probably governed by surface deposition processes. As discussed earlier, we adopt a diurnally averaged surface deposition velocity of 0.1 cm s^{-1} for continental areas in the winter. The surface resistance of freshwater and oceans has been found to be quite large, in the range of $10\text{--}100 \text{ s cm}^{-1}$ [Aldaz, 1969; Galbally and Roy, 1980; Garland et al., 1980; Wesely et al., 1981; Lenschow et al., 1982; Colbeck and Harrison, 1985]. An estimate by Wesely [1983] of the surface resistance for the ocean gives 20 s cm^{-1} under various stability classifications. Taking this estimate, an averaged O_3 deposition velocity over oceanic area can be calculated to be 0.05 cm s^{-1} or less.

The large variability and uncertainty in the surface deposition velocity in the winter make it difficult to estimate the O_3 lifetime due to surface loss. If a deposition velocity of 0.1 cm s^{-1} for the land and 0.05 cm s^{-1} for the ocean is assumed, a lifetime for the whole column O_3 in the mid-latitude of about 150 days is derived. For O_3 in the continental boundary layer, assuming a typical PBL height of 500 m in the winter, the lifetime due to surface deposition is only about 6 days. As discussed above, the question can be raised as how much O_3 or its precursors can be transported out of the boundary layer before they are lost to the surface. The vertical exchange velocity between the PBL and the free troposphere in the winter is probably greater than 0.1 cm s^{-1} , the O_3 deposition velocity. If this is the case, then at least 50% of the O_3 would be transported out of the PBL and hence would be susceptible to long range transport.

Transport of mid-latitude O_3 to the tropics may be an important sink. However, we note that the photochemical O_3 lifetime at 500 mbar at 20° latitude in the winter is as long as 35 days. Therefore mid-latitude O_3 has to be transported to the boundary layer in the tropics to be effectively destroyed. The transport process itself may take substantial time. A model with realistic transport is needed to study this problem.

IMPLICATIONS FOR OZONE DISTRIBUTION

The combination of long O_3 lifetime and the predominance of O_3 production from anthropogenic sources in the winter may have several important implications for the O_3 distribution in the NH in the winter. First, anthropogenic O_3 may be transported over most of the NH. Second, the winter O_3 may be mostly of anthropogenic origin, especially in the lower troposphere of mid- and high latitudes. Furthermore, the long O_3 lifetime allows anthropogenically produced O_3 to accumulate continuously during the winter and to contribute substantially to the observed spring maximum over many remote stations, even as far as Mauna Loa, Hawaii [Oltmans, 1981; Logan, 1985]. Neglect of the Arctic night effect may lead to an overestimate of the winter anthropogenic O_3 production but will have little effect on the spring maximum because it is compensated by the increased production in the spring.

The spring O_3 peak has always been considered to be due to the stratospheric O_3 intrusions [e.g., Junge, 1963; Fabian and Pruchniewicz, 1977; Logan, 1985]. There are several pieces of evidence supporting this theory [see Liu et al., 1980]. The spring O_3 maximum correlates with tracers from the stratosphere such as ^{90}Sr and ^7Be . The maximum in mid-latitudes appears first in the upper troposphere and propagates to the lower troposphere [Chatfield and Harrison, 1977], and a three-dimensional general circulation model (GCM) that included only stratospheric O_3 intrusion and surface deposition suc-

cessfully simulated the spring maximum in the remote areas [Levy et al., 1985]. The model did not include tropospheric photochemistry. The spring O_3 maximum calculated in the model was the result of maximum stratospheric O_3 flux predicted by the model. The long O_3 photochemical lifetime in winter calculated here implies that the relative value of the spring maximum calculated by the GCM would have been substantially greater if the photochemical sink of O_3 was included in the model. In fact, even with constant stratospheric O_3 flux, a spring O_3 maximum would be expected because of the long O_3 lifetime in winter.

By proposing that the anthropogenic O_3 production in the winter contributes substantially to the spring maximum, we do not dispute that the stratospheric intrusion also contributes. In fact, the stratospheric intrusion probably dominates in the upper troposphere. Transport processes like this play an important role in the spatial and temporal distribution of tropospheric O_3 , especially in the winter season when the O_3 lifetime is long. The simple O_3 budget analysis performed above should be regarded as a qualitative assessment. Realistic models are needed to evaluate the relative importance of various O_3 sources.

Our proposal for the O_3 spring maximum is consistent with the recent results by Penkett and Brice [1986]. They used PAN as a tracer of photochemical activity in the troposphere. Based on the observed correlation between PAN and O_3 and the springtime PAN maximum in background air, they suggested that tropospheric photochemistry may contribute to the spring maximum in the tropospheric O_3 concentration.

The proposed dominance of the anthropogenic O_3 source in the winter and its contribution to the spring O_3 maximum provides an interpretation for the long-term variability of O_3 that has been observed in polluted as well as remote areas. We expect that the anthropogenic impact on O_3 will spread over most of the NH in the winter. In contrast, in the summer the impact will probably be confined to the mid-latitudes and may even be confined regionally in the continental boundary layer because of the shortened O_3 lifetime due to surface deposition.

Recently, Oltmans and Komhyr [1986] reported O_3 measurements from 1973 to 1984 at four NOAA Geophysical Monitoring for Climatic Change (GMCC) baseline observatories. They show an increase in O_3 over this period at Mauna Loa, Hawaii (20°N , 155°W , 680 mbar). The linear growth rates in percent per year are $1.97 (\pm 1.04)$, $1.85 (\pm 1.26)$, $0.52 (\pm 1.42)$, and $1.07 (\pm 1.19)$ for winter, spring, summer, and fall seasons, respectively. The numbers in the parentheses are 95% confidence levels of the average values. Only winter and spring seasons have statistically significant growth rates. Because of reduced photochemistry in these seasons the O_3 trend was interpreted by Oltmans and Komhyr [1986] to be due to a change in transport induced by El Niño events. Alternatively, the present results indicate that this trend could be due to increasing O_3 production from anthropogenic emissions of NO_x and NMHC in the winter and spring. The 2% per year increase is consistent with the O_3 increase observed in the winter at 700 and 500 mbar over Hohenpeissenberg, Germany, one of the most consistently operated ozonesonde stations, in about the same period [Logan, 1985]. Other ozonesonde stations in the NH analyzed by Logan [1985] also show positive trends at 700 and 500 mbar. However, the values are significantly lower, and some of them are not statistically significant. We interpret these positive O_3 trends as the result of the increase in NO_x and NMHC emissions in the NH. Unfor-

tunately, sufficiently reliable and accurate emission trends for the NH are not available.

The long-term trend at Mauna Loa and the winter trend at Hohenpeissenberg can be considered as pieces of evidence supporting our proposal. However, the observed O_3 trend at the other GMCC observatory in the NH (i.e., Point Barrow, Alaska) does not. The trend at this site is significant in the summer and fall seasons but not in the winter and spring seasons. The summer and fall growth rate is about the same as Hohenpeissenberg and is consistent with the notion that the site is under the influence of the mid-latitude pollution. The lack of trend in the winter and spring is not consistent with our proposal. Another Arctic station, Resolute (75°N), also shows no trend at 700 and 500 mbar in the winter [Logan, 1985]. A possible explanation is the destruction of O_3 due to anthropogenic emissions of NO, NMHC, and other reducing pollutants in the polar night.

It would be very valuable for testing our proposal if measurements of O_3 could be made at several remote sites like Mauna Loa in the NH, preferably with altitude profiles. Interannual correlation of O_3 between polluted and remote sites at various seasons should show clear differences between summer and winter. We expect good correlation above the boundary layer in the winter and much smaller correlation in the summer, especially in the boundary layer. It would be also useful if existing ozonesonde data for each season could be evaluated for interannual correlations.

SUMMARY AND CONCLUSIONS

From the above discussion it is clear that the tropospheric O_3 budget and distribution is an extremely complex problem that involves photochemical and transport processes of various temporal and spatial scales. It will take considerable efforts in laboratory and field measurements and modeling to understand all the essential aspects of the problem. As mentioned in the introduction, there have been important advances in our understanding of the problem, yet these have almost always been followed by new contradictions and controversies. This study will not be an exception. However, we believe that we have gained some important insights into the budget and distribution of the tropospheric O_3 by analyzing the observed O_3 and NO_x relationship at Niwot Ridge. The highlights are summarized below.

Within a factor of 2, the observed daily ozone increase in the summer can be modeled by photochemical production and destruction plus surface loss. Both model calculations and observations show that the daily O_3 increase per unit of NO_x is greater for lower NO_x . The model calculations without NMHC substantially underestimate the O_3 increase at NO_x higher than about 1.5 ppbv and show the opposite dependence on NO_x . The model calculations with NMHC are reasonably consistent with the observed data, thus supporting the importance of NMHC chemistry in O_3 production.

The summer daily O_3 increases at various NO_x levels at Niwot Ridge have been compared to those from eight other rural stations with concurrent O_3 and NO_x measurements in the central and eastern United States [Research Triangle Institute, 1975; Kelly et al., 1984a]. With only one exception, the daily O_3 increases for these stations agree very well with the O_3 and NO_x relationship observed at Niwot Ridge, a remarkable agreement considering the wide range of geographical locations. The consistency of the summer daily O_3 increases

suggests that the average daily O_3 production at a rural station may be predicted if the NO_x concentration is known. The dependence of the O_3 production rate on NO_x also allows us to formulate an approximate method to estimate the O_3 production from NO_x and NMHC emissions. The method uses the concept that the O_3 production is proportional to the NO_x emission rate and its lifetime.

The method outlined here provides new insight into some of the important problems of the tropospheric O_3 budget and distribution. It is shown that most of the O_3 due to human activities is probably produced from the interaction of anthropogenic NO_x with NMHC. The contribution from CO and CH_4 is minor, especially in summer. In addition, photochemistry and transport of NMHC and their products such as PAN play such a critical role in the interaction of CO and CH_4 with anthropogenic NO_x that previous evaluations of O_3 production from this interaction need to be reevaluated.

For the United States we estimate an average summer column O_3 production rate due to anthropogenic NO_x and NMHC emissions of about $1 \times 10^{12} \text{ cm}^{-2} \text{ s}^{-1}$, about 20 times the average cross-tropopause O_3 flux. Estimates of O_3 production from natural NO_x sources range from 1.9×10^{11} to $12 \times 10^{11} \text{ cm}^{-2} \text{ s}^{-1}$. Therefore human activities probably contribute 50–80% of the O_3 in the central and eastern United States in the summer. The environmental effects due to the increased O_3 on crops and forest may be substantial [Heck et al., 1982; Adams et al., 1985; Reich and Amundson, 1985]. A similar situation is expected to exist in Europe.

Averaged over the NH, the anthropogenic O_3 production in the summer is about $1 \times 10^{11} \text{ cm}^{-2} \text{ s}^{-1}$. The production of O_3 from natural NO_x emissions is greater, roughly $3 \times 10^{11} \text{ cm}^{-2} \text{ s}^{-1}$ in the summer. Both are greater than the cross-tropopause O_3 flux. Because the O_3 lifetime is relatively short in the summer, especially in the PBL, the O_3 distribution is probably controlled by regional sources.

The winter daily O_3 production rate is of the order of 10% of the summer value at the same NO_x level. However, because the NO_x lifetime is about 10 times longer when only daytime chemistry is considered, the O_3 production rate integrated over the lifetime of NO_x in the winter is comparable to the summer value. Since the natural NO_x sources are insignificant compared to the anthropogenic source in the winter, the O_3 budget in the NH should be dominated by the latter. In this connection it should be noted that the long lifetime of PAN and NO_x in the winter may allow a significant export of anthropogenic NO_x to remote regions. The dilution of this NO_x coupled with the nonlinear dependence of production on NO_x may significantly increase the ozone production efficiency from anthropogenic NO_x emissions during the winter season.

The photochemical lifetime of O_3 in mid-latitudes in the winter is of the order of 200 days. We propose that accumulation of anthropogenically produced O_3 may contribute substantially to the observed spring O_3 maximum in the lower troposphere of the NH, a phenomenon that has often been considered to be due to enhanced stratosphere-troposphere exchange. In addition, the long lifetime will allow transport of O_3 not only zonally but also to other latitudes. It is proposed that the observed long-term O_3 trend in winter and spring seasons at Mauna Loa, Hawaii, a clean site, and at Hohenpeissenberg, Germany, a moderately polluted site, may be due to increases in the same anthropogenic source.

The major uncertainties in the winter O_3 budget and distri-

bution is associated with the estimates of lifetimes for NO_x and O_3 . These involve the nighttime NO_3 and N_2O_5 removal mechanism, surface deposition of NO_x and O_3 , and removal of secondary NMHC products such as PAN and aldehydes. The photochemistry of NO_3 , N_2O_5 , and the organic nitrates is not well understood. Laboratory studies of the photochemistry of these species and reactions of NMHC and NO_x in general are needed. Since transport processes play an important role in the O_3 production efficiency and the fate of organic nitrates, models with realistic transport parameterization will be needed to address the complexities of coupled chemistry and dynamics. Finally, measurements of O_3 and its precursors, especially in the remote troposphere, will be most valuable to improve our knowledge of the O_3 budget and distribution.

Acknowledgments. This research has been funded as part of the National Acid Precipitation Assessment Program by the National Oceanic and Atmospheric Administration. The paper has benefited from preprints sent by J. Fishman, and by S. J. Oltmans and W. K. Komhyr. Discussions with D. L. Albritton, M. A. Carroll, J. P. Greenberg, C. J. Howard, D. H. Lenschow, B. A. Ridley, A. R. Ravishankara, S. Solomon, and P. R. Zimmerman have been helpful. Programming help from R. Tallamraju is appreciated.

REFERENCES

- Adams, R. M., S. A. Hamilton, and B. A. McCarl, An assessment of the economic effects of ozone on U.S. agriculture, *J. Air Pollut. Control Assoc.*, **35**, 938–943, 1985.
- Albritton, D. L., S. C. Liu, and D. Kley, Global nitrate deposition from lightning, in *Environmental Impact of Natural Emissions*, edited by V. P. Aneja, pp. 100–112, Air Pollution Control Association, Pittsburgh, Pa., 1984.
- Aldaz, L., Flux measurements of atmospheric ozone over land and water, *J. Geophys. Res.*, **74**, 6943–6946, 1969.
- Altshuller, H. P., The role of nitrogen oxides in nonurban ozone formation in the planetary boundary layer over N. America, W. Europe and adjacent areas of ocean, *Atmos. Environ.*, **20**, 245–268, 1986.
- Atkinson, R., and A. C. Lloyd, Evaluation of kinetic and mechanistic data for modeling of photochemical smog, *J. Phys. Chem. Ref. Data*, **13**, 315–444, 1984.
- Atkinson, R., A. C. Lloyd, and L. Wings, An updated chemical mechanism for hydrocarbon/ NO_x / SO_2 photooxidations suitable for inclusion in atmospheric simulation models, *Atmos. Environ.*, **16**, 1341–1355, 1982.
- Atkinson, R., C. N. Plum, W. P. L. Carter, A. M. Winer, and J. N. Pitts, Jr., Rate constants for the gas-phase reactions of nitrate radicals with a series of organics in air at 298 ± 1 K, *J. Phys. Chem.*, **88**, 1210–1215, 1984.
- Barrie, L. A., and R. M. Hoff, Five years of air chemistry observations in the Canadian arctic, *Atmos. Environ.*, **19**, 1995–2010, 1985.
- Barrie, L. A., R. M. Hoff, and S. M. Daggupaty, The influence of mid-latitude pollution sources on haze in the Canadian arctic, *Atmos. Environ.*, **15**, 1407–1420, 1981.
- Bottger, A., G. Gravenhorst, and D. H. Ehhalt, Atmosphärische Kreisläufe von Stickoxiden und Ammoniak, *Ber. Kernforschungsanlage Jülich*, **1558**, 95, 1978.
- Chameides, W. L., and A. Tan, The two-dimensional diagnostic model for tropospheric OH: An uncertainty analysis, *J. Geophys. Res.*, **86**, 5209–5223, 1981.
- Chameides, W. L., and J. C. G. Walker, A photochemical theory of tropospheric ozone, *J. Geophys. Res.*, **78**, 8751–8760, 1973.
- Chang, T. Y., J. M. Norbeck, and B. Weinstock, An estimate of the NO_x removal rate in an urban atmosphere, *Environ. Sci. Technol.*, **13**, 1534–1537, 1979.
- Chatfield, R., and H. Harrison, Tropospheric ozone, 2, Variations along a meridional band, *J. Geophys. Res.*, **82**, 5969–5976, 1977.
- Cleveland, W. S., B. Kleiner, J. E. McRae, J. L. Warner, and R. E. Pasceri, Geographical properties of ozone concentrations in the northeastern United States, *J. Air Pollut. Control Assoc.*, **27**, 325–328, 1977.
- Colbeck, I., and R. M. Harrison, Dry deposition of ozone: Some measurements of deposition velocity and of vertical profiles to 100 meters, *Atmos. Environ.*, **19**, 1807–1818, 1985.
- Costanza, J., and J. H. Seinfeld, Optimal emission control strategies for photochemical smog, *Environ. Sci. Technol.*, **16**, 98–102, 1982.
- Cox, R. A., A. E. J. Eggleton, R. G. Derwent, J. E. Lovelock, and D. E. Pack, Long-range transport of photochemical ozone in north-western Europe, *Nature*, **255**, 118–121, 1975.
- Crutzen, P. J., A discussion of the chemistry of some minor constituents in the stratosphere and troposphere, *Pure Appl. Geophys.*, **106**, 1385–1399, 1973.
- Crutzen, P. J., and L. T. Gidel, A two-dimensional photochemical model of the atmosphere, 2, The tropospheric budgets of the anthropogenic chlorocarbons, CO , CH_4 , CH_3Cl and the effect of various NO_x sources on tropospheric ozone, *J. Geophys. Res.*, **88**, 6641–6661, 1983.
- Danielsen, E. F., and V. A. Mohnen, Project dust storm report: Ozone transport, in situ measurements, and meteorological analyses of tropopause folding, *J. Geophys. Res.*, **82**, 5867–5877, 1977.
- Dütsch, H. U., W. Züllig, and C. Ling, Regular ozone observation at Thalwil, Switzerland and at Boulder, Colorado, *Rep. LAPETH-I, Lab. für Atmos. Eidgenöss. Tech. Hochschule, Zurich*, 1970.
- Eichmann, R., P. Neuling, G. Ketseridis, J. Hahn, R. Jaenicke, and C. Junge, N-alkane studies in the troposphere, I, Gas and particulate concentrations in North Atlantic air, *Atmos. Environ.*, **13**, 587–599, 1979.
- Eichmann, R., G. Ketseridis, G. Schebeske, R. Jaenicke, J. Hahn, P. Warneck, and C. Jung, N-alkane studies in the troposphere, II, Gas and particulate concentrations in Indian Ocean air, *Atmos. Environ.*, **14**, 695–703, 1980.
- Fabian, P., and P. G. Pruchniewicz, Meridional distribution of ozone in the troposphere and its seasonal variations, *J. Geophys. Res.*, **82**, 2063–2073, 1977.
- Fahey, D. W., G. Hubler, D. D. Parrish, E. J. Williams, R. E. Norton, B. A. Ridley, H. B. Singh, S. C. Liu, and F. C. Fehsenfeld, Reactive nitrogen species in the troposphere: Measurements of NO , NO_2 , HNO_3 , particulate nitrate, peroxyacetyl nitrate (PAN), O_3 , and total reactive odd nitrogen (NO_y) at Niwot Ridge, Colorado, *J. Geophys. Res.*, **91**, 9781–9793, 1986.
- Fehsenfeld, F. C., M. J. Bollinger, S. C. Liu, D. D. Parrish, M. McFarland, M. Trainer, D. Kley, P. C. Murphy, D. L. Albritton, and D. H. Lenschow, A study of ozone in the Colorado Mountains, *J. Atmos. Chem.*, **1**, 87–105, 1983.
- Ferman, M. A., G. T. Wolff, and N. A. Kelly, The nature and sources of haze in the Shenandoah Valley/Blue Ridge Mountains area, *J. Air Pollut. Control Assoc.*, **31**, 1074–1082, 1981.
- Fishman, J., and W. Seiler, Correlative nature of ozone and carbon monoxide in the troposphere: Implications for the tropospheric ozone budget, *J. Geophys. Res.*, **88**, 3662–3670, 1983.
- Fishman, J., S. Solomon, and P. J. Crutzen, Observational and theoretical evidence in support of a significant in-situ photochemical source of tropospheric ozone, *Tellus*, **31**, 432–446, 1979.
- Fishman, J., F. M. Vukovich, and E. V. Browell, The photochemistry of synoptic-scale ozone synthesis: Implications for the global tropospheric ozone budget, *J. Atmos. Chem.*, **3**, 299–320, 1985.
- Galbally, I. E., and C. R. Roy, Loss of fixed nitrogen from soils by nitric oxide exhalation, *Nature*, **275**, 734–735, 1978.
- Galbally, I. E., and C. R. Roy, Destruction of ozone at the earth's surface, *Q. J. R. Meteorol. Soc.*, **106**, 599–620, 1980.
- Garland, J. A., A. W. Elzerman, and S. A. Penkett, The mechanism for dry deposition of ozone to water surfaces, *J. Geophys. Res.*, **85**, 7488, 7492, 1980.
- Greenberg, J. P., and P. R. Zimmerman, Nonmethane hydrocarbons in remote tropical, continental, and marine atmospheres, *J. Geophys. Res.*, **89**, 4767–4778, 1984.
- Grosjean, D., and J. Harrison, Response of chemiluminescence NO_x analyzers and ultraviolet ozone analyzers to organic air pollutants, *Environ. Sci. Technol.*, **19**, 862–865, 1985.
- Guicherit, R., and H. Van Dop, Photochemical production of ozone in Western Europe (1971–1975) and its relation to meteorology, *Atmos. Environ.*, **11**, 145–156, 1977.
- Heck, W. W., O. C. Taylor, R. M. Adams, G. Bingham, J. Miller, E. Preston, and L. Weinstein, Assessment of crop loss from ozone, *J. Air Pollut. Control Assoc.*, **32**, 353–361, 1982.
- Heintzenberg, J., H. C. Hanssen, and H. Lannfors, The chemical

- composition of arctic haze at Ny-Alesund, Spitsbergen, *Tellus*, **33**, 162–171, 1981.
- Helas, G., and P. Warneck, Background NO_x mixing ratios in air masses over the North Atlantic Ocean, *J. Geophys. Res.*, **86**, 7283–7290, 1981.
- Hoell, J. M., Jr., G. L. Gregory, D. S. McDougal, G. W. Sachse, G. F. Hill, E. P. Condon, and R. A. Rasmussen, An intercomparison of carbon monoxide measurement techniques, *J. Geophys. Res.*, **90**, 12,881–12,889, 1985.
- Hov, O., Ozone in the troposphere: High level pollution, *Ambio*, **13**, 73–79, 1984.
- Hov, O., and R. G. Derwent, Sensitivity studies of the effects of model formulation on the evaluation of control strategies for photochemical air pollution formation in the United Kingdom, *J. Air Pollut. Control Assoc.*, **31**, 1260–1267, 1981.
- Hov, O., J. Schjoldager, and B. M. Wathne, Measurement and modeling of concentrations of terpenes in coniferous forest air, *J. Geophys. Res.*, **88**, 10,679–10,688, 1983.
- Huebert, B. J., and C. H. Robert, The dry deposition of nitric acid to grass, *J. Geophys. Res.*, **90**, 2085–2090, 1985.
- Isaksen, I. S. A., O. Hov, S. A. Penkett, and A. Semb, Model analysis of the measured concentration of organic gases in the Norwegian arctic, *J. Atmos. Chem.*, **3**, 3–27, 1985.
- Jet Propulsion Laboratory, Chemical kinetics and photochemical data for use in stratospheric modeling, *Eval. 6*, Calif. Inst. of Technol., Pasadena, 1985.
- Judeikis, H. S., and A. G. Wren, Laboratory measurements of NO and NO_2 deposition onto soil and cement surfaces, *Atmos. Environ.*, **12**, 2315–2319, 1978.
- Junge, C. E., *Air Chemistry and Radioactivity*, Academic Orlando, Fla., 1963.
- Junge, C. E., W. Seiler, and P. Warneck, The atmospheric ^{12}CO and ^{14}CO budget, *J. Geophys. Res.*, **76**, 2866–2879, 1971.
- Kaimal, J. C., J. C. Wyngaard, D. A. Haugen, O. R. Coté, Y. I. Zumi, S. J. Caughey, and C. J. Readings, Turbulence structure in the convective boundary layer, *J. Atmos. Sci.*, **76**, 2152–2169, 1976.
- Kasting, J. F., and H. B. Singh, Photochemistry of nonmethane hydrocarbons in the troposphere, *Eos Trans. AGU*, **66**, 823, 1985.
- Kelly, N. A., G. T. Wolff, and M. A. Ferman, Sources and sinks of ozone in rural areas, *Atmos. Environ.*, **18**, 1251–1266, 1984a.
- Kelly, T. J., R. L. Tanner, L. Newman, P. J. Galvin, and J. A. Kadlec, Trace gas and aerosol measurements at a remote site in the northeast U.S., *Atmos. Environ.*, **18**, 2565–2576, 1984b.
- Kircher, C. C., and S. P. Sander, Kinetics and mechanism of HO_2 and DO_2 disproportionations, *J. Phys. Chem.*, **88**, 2082–2091, 1984.
- Kley, D., J. W. Drummond, M. McFarland, and S. C. Liu, Tropospheric profiles of NO_x , *J. Geophys. Res.*, **86**, 3153–3161, 1981.
- Lee, Y. N., and S. E. Schwartz, Evaluation of the rate of uptake of nitrogen dioxide by atmospheric and surface liquid water, *J. Geophys. Res.*, **86**, 11,971–11,984, 1981.
- Lenschow, D. H., R. Pearson, Jr., and B. B. Stankor, Measurements of ozone vertical flux to ocean and forest, *J. Geophys. Res.*, **87**, 8833–8837, 1982.
- Levy, H., II, Normal atmosphere: Large radical and formaldehyde concentrations predicted, *Science*, **173**, 141–143, 1971.
- Levy, H., II, J. D. Mahlman, W. J. Moxim, and S. C. Liu, Tropospheric ozone: The role of transport, *J. Geophys. Res.*, **90**, 3753–3772, 1985.
- Liu, S. C., Possible effects on tropospheric O_3 and OH due to NO emissions, *Geophys. Res. Lett.*, **4**, 325–328, 1977.
- Liu, S. C., D. Kley, M. McFarland, J. D. Mahlman, and H. Levy II, On the origin of tropospheric ozone, *J. Geophys. Res.*, **85**, 7546–7552, 1980.
- Liu, S. C., M. McFarland, D. Kley, O. Zafiriou, and B. Huebert, Tropospheric NO_x and O_3 budgets in the equatorial Pacific, *J. Geophys. Res.*, **88**, 1360–1368, 1983.
- Lloyd, A. C., R. Atkinson, F. W. Lurmann, and B. Nitta, Modeling potential ozone impacts from natural hydrocarbons, I, Development and testing of a chemical mechanism for the NO_x -air photooxidations of isoprene and α -pinene under ambient conditions, *Atmos. Environ.*, **18**, 1133–1143, 1984.
- Logan, J. A., Nitrogen oxides in the troposphere: Global and regional budgets, *J. Geophys. Res.*, **88**, 10,785–10,807, 1983.
- Logan, J. A., Tropospheric ozone: Seasonal behavior, trends and anthropogenic influence, *J. Geophys. Res.*, **90**, 10,463–10,482, 1985.
- Logan, J. A., M. J. Prather, S. C. Wofsy, and M. B. McElroy, Tropospheric chemistry: A global perspective, *J. Geophys. Res.*, **86**, 7210–7254, 1981.
- Mahlman, J. D., H. Levy II, and W. J. Moxim, Three-dimensional tracer structure and behavior as simulated in two ozone precursor experiments, *J. Atmos. Sci.*, **37**, 655–685, 1980.
- Martinez, J. R., and H. B. Singh, Survey of the role of NO_x in nonurban ozone formation, *Rep. Proj. 6780-8*, SRI Int., Menlo Park, Calif., 1979.
- McFarland, M., D. Kley, J. W. Drummond, A. L. Schmeltekopf, and R. H. Winkler, Nitric oxide measurements in the equatorial Pacific region, *Geophys. Res. Lett.*, **6**, 605–608, 1979.
- Morris, E. D., and H. Niki, Reaction of the nitrate radical with acetaldehyde and propylene, *J. Phys. Chem.*, **78**, 1337–1338, 1974.
- Mueller, P. K., and G. M. Hidy, The sulfate regional experiment (SURE): Report of findings, *Rep. EA-1901*, Electr. Power Res. Inst., Palo Alto, Calif., March 1983.
- Noxon, J. F., Tropospheric NO_2 , *J. Geophys. Res.*, **83**, 3051–3057, 1978.
- Noxon, J. F., NO_3 and NO_2 in the mid-Pacific troposphere, *J. Geophys. Res.*, **88**, 11,017–11,021, 1983.
- Oltmans, S. J., Surface ozone measurements in clean air, *J. Geophys. Res.*, **86**, 1174–1180, 1981.
- Oltmans, S. J., and W. D. Komhyr, Surface ozone distributions and variations from 1973–1984 Measurements at the NOAA Geophysical Monitoring for Climatic Change baseline observatories, *J. Geophys. Res.*, **91**, 5229–5236, 1986.
- Oort, A. H., Global atmospheric circulation statistics, 1958–1973, *NOAA Prof. Pap. 14*, NOAA, Dep. of Commer., Rockville, Md., 1983.
- Parrish, D. D., P. C. Murphy, D. L. Albritton, and F. C. Fehsenfeld, The measurement of the photodissociation rate of NO_2 in the atmosphere, *Atmos. Environ.*, **17**, 1365–1379, 1983.
- Parrish, D. D., D. W. Fahey, E. J. Williams, S. C. Liu, M. Trainer, P. C. Murphy, D. L. Albritton, and F. C. Fehsenfeld, Background ozone and anthropogenic ozone enhancement at Niwot Ridge, Colorado, *J. Atmos. Chem.*, **4**, 63–80, 1986a.
- Parrish, D. D., B. Huebert, R. B. Norton, M. J. Bollinger, S. C. Liu, P. C. Murphy, D. L. Albritton, and F. C. Fehsenfeld, Measurements of HNO_3 and NO_3^- particulates at a rural site in the Colorado Mountains, *J. Geophys. Res.*, **91**, 5379–5393, 1986b.
- Penkett, S. A., and K. A. Brice, The spring maximum in photo-oxidants in the northern hemisphere troposphere, *Nature*, **319**, 655–658, 1986.
- Platt, U., A. M. Winer, H. W. Biermann, R. Atkinson, and J. N. Pitts, Jr., Measurement of nitrate radical concentrations in continental air, *Environ. Sci. Technol.*, **18**, 365–369, 1984.
- Pratt, R., and P. Falconer, Circumpolar measurements of ozone and carbon monoxide from a commercial airliner, *J. Geophys. Res.*, **84**, 7876–7882, 1979.
- Rahn, K. A., and R. J. McCaffrey, Long-range transport of pollution aerosol to the arctic: A problem without borders, paper presented at WMO Symposium on the Long-Range Transport of Pollutants, World Meteorol. Organ., Sofia, 1979.
- Reich, P. B., and R. G. Amundson, Ambient levels of ozone reduce net photosynthesis in tree and crop species, *Science*, **230**, 566–570, 1985.
- Research Triangle Institute, Investigation of rural oxidant levels as related to urban hydrocarbon control strategies, *Rep. EPA-450/3-75-036*, 359 pp., Environ. Prot. Agency, Research Triangle Park, N. C., 1975.
- Ridley, B. A., M. A. Carroll, and G. L. Gregory, Measurements of nitric oxide in the boundary layer and free troposphere over the Pacific Ocean, *J. Geophys. Res.*, **92**, 2025–2048, 1987.
- Roberts, J. M., F. C. Fehsenfeld, D. L. Albritton, and R. E. Sievers, Measurements of monoterpene hydrocarbons at Niwot Ridge, Colorado, *J. Geophys. Res.*, **88**, 10,667–10,678, 1983.
- Roberts, J. M., F. C. Fehsenfeld, S. C. Liu, M. J. Bollinger, C. Hahn, D. L. Albritton, and R. E. Sievers, Measurements of aromatic hydrocarbon ratios and NO_x concentrations in the rural troposphere: Observation of air mass photochemical aging and NO_x removal, *Atmos. Environ.*, **18**, 2421–2432, 1984.
- Roberts, J. M., R. S. Hutte, F. C. Fehsenfeld, D. L. Albritton, and R. E. Sievers, Measurement of anthropogenic hydrocarbon concentration ratios in the rural troposphere: Discrimination between background and urban sources, *Atmos. Environ.*, **19**, 1945–1950, 1985.

- Rogers, H. H., H. E., Jeffries, E. P. Stahel, W. W. Heck, L. A. Ripperton, and A. M. Witherspoon, Measuring air pollutant uptake by plants: A direct kinetic technique, *J. Air Pollut. Control Assoc.*, 27, 1192–1197, 1977.
- Rudolph, J., and D. H. Ehhalt, Measurements of C_2 – C_5 hydrocarbons over the North Atlantic, *J. Geophys. Res.*, 86, 11,959–11,964, 1981.
- Sakamaki, F., M. Okuda, H. Akimoto, and H. Yamazaki, Computer modeling study of photochemical ozone formation in the propane-nitrogen oxides-dry air system, generalized maximum ozone isopleth, *Environ. Sci. Technol.*, 16, 45–52, 1982.
- Schiff, H. I., D. Pepper, and B. A. Ridley, Tropospheric NO measurements up to 7 km, *J. Geophys. Res.*, 84, 7895–7897, 1979.
- Seiler, W., and P. Crutzen, Estimates of the gross and net flux of carbon between the biosphere and the atmosphere from biomass burning, *Clim. Change*, 2, 207–247, 1980.
- Shaw, R. W., and R. J. Paur, Measurement of sulfur in gases and particles during sixteen months in the Ohio River Valley, *Atmos. Environ.*, 17, 1431–1438, 1983.
- Singh, H. B., L. J. Salas, B. A. Ridley, J. D. Shetter, N. M. Donahue, F. C. Fehsenfeld, D. W. Fahey, D. D. Parrish, E. J. Williams, S. C. Liu, G. Hubler, and P. C. Murphy, Relationship between peroxyacetyl nitrate and nitrogen oxides in the clean troposphere, *Nature*, 318, 347–349, 1985.
- Slemr, F., and W. Seiler, Field measurements of NO and NO₂ emissions from fertilized and unfertilized soils, *J. Atmos. Chem.*, 2, 1–24, 1984.
- Spicer, C. W., D. W. Joseph, P. R. Stickse, and G. R. Ward, Ozone sources and transport into the northeastern United States, *Environ. Sci. Technol.*, 13, 975–985, 1979.
- Trainer, M., E. Y. Hsie, S. H. McKeen, R. Tallamraju, D. D. Parrish, D. W. Fahey, F. C. Fehsenfeld, and S. C. Liu, Model study of peroxy radical concentrations within the planetary boundary layer, *J. Geophys. Res.*, in press, 1987.
- Turman, B. N., and B. C. Edgar, Global lightning distribution at dawn and dusk, *J. Geophys. Res.*, 87, 1191–1206, 1982.
- U.S. Environmental Protection Agency, Uses, limitations and technical basis of procedures for quantifying relationships between photochemical oxidants and precursors, *Rep. EPA 450/2-77-021a*, Research Triangle Park, N. C., 1977.
- U.S. Environmental Protection Agency, National air pollutant emission estimates, 1940–1980, *Rep. EPA-450/4-82-001*, Research Triangle Park, N. C., 1982.
- Vukovich, F. M., W. D. Bach, Jr., B. W. Crissman, and W. J. King, On the relationship between high ozone in the rural surface layer and high pressure systems, *Atmos. Environ.*, 11, 967–984, 1977.
- Vukovich, F. M., J. Fishman and E. V. Browell, The reservoir of ozone in the boundary layer of the eastern United States and its potential impact on the global ozone budget, *J. Geophys. Res.*, 90, 5687–5698, 1985.
- Wesely, M. L., Turbulent transport of ozone to surfaces common in the eastern half of the United States, in *Trace Atmospheric Constituents*, edited by S. E. Schwartz, pp. 345–370, John Wiley, New York, 1983.
- Wesely, M. L., D. R. Cook, and R. M. Williams, Field measurement of small ozone fluxes to snow, wet bare soil, and lake water, *Boundary Layer Meteorol.*, 20, 459–471, 1981.
- Wesely, M. L., J. A. Eastman, D. H. Stedman, and E. D. Yalvac, An eddy correlation measurement of NO₂ flux to vegetation and comparison to O₃ flux, *Atmos. Environ.*, 16, 815–820, 1982.
- Williams, E. J., D. W. Fahey, G. Hübler, D. D. Parrish, P. C. Murphy, R. B. Norton, C. S. Eubank, D. L. Albritton, and F. C. Fehsenfeld, Measurements of NO, NO₂ and O₃ at Niwot Ridge, Colorado, *Eos Trans. AGU*, 65, 833, 1984.
- Williams, E. J., D. D. Parrish, D. L. Albritton, and F. C. Fehsenfeld, Measurements of NO_x fluxes from soil: Results of enclosure measurements, *Eos Trans. AGU*, 66, 819, 1985.
- Wolff, G. T., and P. J. Lioy, Development of an ozone river associated with synoptic scale episodes in the eastern United States, *Environ. Sci. Technol.*, 14, 1257–1260, 1980.
- D. W. Fahey, F. C. Fehsenfeld, G. Hübler, S. C. Liu, P. C. Murphy, D. D. Parrish, M. Trainer, and E. J. Williams, Environmental Research Laboratories, NOAA, 325 Broadway, Boulder, CO 80309.

(Received May 1, 1986;
revised January 7, 1987;
accepted January 13, 1987.)